SATASHKENT
College Prep Community

math book 3.0

August, September, October 2025 questions

You will never walk alone...!

Contents

Introduction

Welcome to the MathBook by @satashkent.

Many real exam questions are officially and unofficially shared across the Internet, providing valuable practice materials for SAT students. However, these resources are often scattered, unstructured, and not categorized by topic, making it difficult for students to focus on specific areas of improvement.

Recognizing this challenge, our team created **MathBook**—a structured collection of **real exam questions from the past years, categorized by topic.** This book is designed for all **SAT learners** who want an efficient and organized approach to mastering Digital SAT Math. We have dedicated countless hours to compiling and structuring these questions to ensure you get the most relevant and targeted practice.

This book is divided into topics, covering all question types that appear on the DSAT. Each topic includes carefully selected real exam-level questions along with an answer sheet to help you track your progress, analyze mistakes, and strengthen weak areas. This structured approach will boost your confidence and help you recognize patterns in SAT Math problems.

Our **primary goal** is to **support students who are committed to excelling in SAT Math.**

Let's master SAT Math together!

How to Use This Book?

Step 1: **Open the book.** If you've made it, congrats!

Step 2: **Pick a question.** Go in order, choose a random one, or let ChatGPT choose.

Step 3: **Solve it.** Grab a pencil, solve, and try not to overthink. If you get stuck, stare at the "Patalok" for a while.

Step 4: **Check your answer.** If it' s right, go next, no time for Celebrating. If it's wrong, try again.

Step 5: **Repeat!** Math skills don't grow on trees (unfortunately).

References

All questions in this book are from College Board Real Exam Questions (2023–2025). This book is not affiliated with or endorsed by the College Board and is for educational use only.

Algebra

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Algebra Formulas

- **Integer -** A whole number, not a fraction.
- **Domain (x) -** The set of possible x values.
- **Range (y) -** The set of possible y values.
- **x-intercept -** Where $y = 0$.
- **y-intercept -** Where $x = 0$.
- **Standard form:**

$$a_1x + b_1y = c_1$$
$$a_2x + b_2y = c_2$$

One solution	$left_1 \neq left_2$	$\frac{a_1}{a_2} \neq \frac{b_1}{b_2}$
No solution	$left_1 = left_2$ $right_1 \neq right_2$	$\frac{a_1}{a_2} = \frac{b_1}{b_2} \neq \frac{c_1}{c_2}$
∞ solution	$left_1 = left_2$ $right_1 = right_2$	$\frac{a_1}{a_2} = \frac{b_1}{b_2} = \frac{c_1}{c_2}$

- **Slope-intercept form:** $y = mx + b$
  - m - slope of the line,
  - b - y-intercept (when $x = 0$),
  - $m = \frac{y_2 - y_1}{x_2 - x_1} = \frac{\text{rise}}{\text{run}}$.

One solution	$m_1 \neq m_2$
No solution	$m_1 = m_2$ $b_1 \neq b_2$
∞ solution	$m_1 = m_2$ $b_1 = b_2$

- **Constant/Coefficient:**
  X should be displayed/shown as constant or coefficient = **X** should be visible in the equation/function itself.

Expressions

1. [August 2025]

Which expression is equivalent to $7x(x+4)$?

A) $7x^2+4$

B) $7x^2+28x$

C) $8x^2+4$

D) $8x^2+11x$

2. [August 2025]

$$65=\frac{2K}{v^2}$$

For an object with a mass of 65 kilograms, the given equation relates the kinetic energy K, in joules, of the object to the object's speed v, in meters per second, where K and v are positive. Which equation correctly expresses this object's speed, in meters per second, in terms of the object's kinetic energy, in joules?

A) $v=\frac{2K}{65^2}$

B) $v=\frac{65^2}{2K}$

C) $v=\sqrt{\frac{65}{2k}}$

D) $v=\sqrt{\frac{2K}{65}}$

3. [August 2025]

The percent increase in mass of a certain red kangaroo from 110 days old to 210 days old was 773%. If this red kangaroo's mass was k grams at 110 days old, which expression represents its mass, in grams, at 210 days old?

A) $0.07k$

B) $1.07k$

C) $7.73k$

D) $8.73k$

4. [August 2025]

The percent increase in mass of a certain red kangaroo from 90 days old to 160 days old was 438%. If this red kangaroo's mass was k grams at 90 days old, which expression represents its mass, in grams, at 160 days old?

A) $0.06k$

B) $1.06k$

C) $4.38k$

D) $5.38k$

5. [August 2025]

Which expression is equivalent to $(8nr^2+2nr)+(4n^2r+3nr)$?

A) $12n^2r^2+5nr$

B) $12n^3r^3+5n^2r^2$

C) $32n^3r^3+6n^2r^2$

D) $8nr^2+4n^2r+5nr$

6. [August 2025]

$$R=\frac{0.43l}{A}$$

The given equation relates the resistance R, in ohms, of a wire to its length l, in meters, and its cross-sectional area A, in square meters. Which equation correctly expresses the length, in meters, of the wire in terms of its resistance, in ohms, and its cross-sectional area, in square meters?

A) $l=\frac{R-A}{0.43}$

B) $l=\frac{R+A}{0.43}$

C) $l=\frac{0.43R}{A}$

D) $l=\frac{AR}{0.43}$

7. [August 2025]

Which expression is equivalent to $6x(x+7)$?

A) $6x^2 + 7x$

B) $6x^2 + 42x$

C) $7x^2 + 7x$

D) $6x^2 + 13x$

8. [August 2025]

$$b - 38 = \frac{x}{y}$$

The given equation relates the positive numbers $b, x,$ and y. Which equation correctly expresses x in terms of b and y?

A) $x = by - 38$

B) $x = \frac{b-38}{y}$

C) $x = by - 38y$

D) $x = \frac{by-38}{y}$

9. [August 2025]

Which expression represents the product of $(x^{-9}y^8z^4)$ and $(x^5z^4 + y^3z^{-7})$?

A) $x^{-4}y^8z^8 + y^3z^{-7}$

B) $4^{-4}z^8 + x^{-9}z^{-3}$

C) $x^{-4}y^8z^8 + x^{-9}y^{11}z^{-3}$

D) $x^{-4}z^8 + y^{11}z^{-3}$

10. [August 2025]

$$\frac{a}{(b+c)} = 55$$

The given equation relates the positive numbers $a, b,$ and c. Which equation correctly expresses a in terms of b and c?

A) $a = (b+c)$

B) $a = 55 - (b+c)$

C) $a = 55(b+c)$

D) $a = \frac{55}{(b+c)}$

11. [September 2025]

$$\frac{a}{b+c} = 64$$

The given equation relates the positive numbers a, b, and c. Which equation correctly expresses a in terms of b and c?

A) $a = b + c$

B) $a = 64 - (b+c)$

C) $a = 64(b+c)$

D) $a = \frac{64}{(b+c)}$

12. [September 2025]

Which expression is equivalent to $(3x^3 - 8x + 5) - (4x^6 + 9x - 2)$?

A) $-4x^6 + 3x^3 - 17x + 7$

B) $-x^9 - 17x - 7$

C) $-x^9 - x + 3$

D) $-4x^6 + 3x^3 + x + 3$

13. [September 2025]

Which expression is equivalent to $v^4 - 590v^3$?

A) $v^3(v + 590)$
B) $(c^2 + 59)(v^2 - 10)$
C) $(v^2 - 59)(v^2 - 10)$
D) $v^3(v - 590)$

14. [September 2025]

$$b - 19 = \frac{x}{y}$$

The given equation relates the positive numbers b, x, and y. Which equation correctly expresses x in terms of b and y?

A) $x = \frac{(b-19)}{y}$
B) $x = by - 19y$
C) $x = \frac{(by-19)}{y}$
D) $x = by - 19$

15. [September 2025]

The length of a rectangle is 35 centimeters greater than its width. Which equation gives the area A, in square centimeters, of the rectangle in terms of the width w, in centimeters, of the rectangle?

A) $A = (w)(w + 35)$
B) $A = (w)(35w)$
C) $A = (w)(\frac{w}{35})$
D) $A = (w)(w - 35)$

16. [September 2025]

Which expression represents the product of $(x^{-9}y^6z^5)$ and $(x^4z^5 + y^8z^{-7})$?

A) $x^{-5}y^6z^{10} + y^8z^{-7}$
B) $x^{-5}y^6z^{10} + x^{-9}y^14z^{-2}$
C) $x^{-5}z^{10} + x^{-9}z^{-2}$
D) $x^{-5}z^{10} + y^14z^{-2}$

17. [October 2025]

$$18qrt - 2qrs + 10rst = 0$$

In the given equation, $q, r, s,$ and t are positive, and $\frac{q}{t}$ is greater than 5. Which expression is equivalent to s?

A) $\dfrac{-9qt}{q - 5t}$
B) $\dfrac{9qt}{q - 5t}$
C) $18qrt$
D) $\dfrac{2qt}{q - 5t}$

18. [October 2025]

If $5c = 15$, what is the value of $5c - 8$?

A) 3
B) 67
C) 7
D) 23

19. [October 2025]

A city has an area of x square miles and the population density of 1035 people per square mile. The suburb adjacent to the city has an area of y square miles and the population density of 535 people per square mile. The combined population of the city and the suburb adjacent to the city is 13,165 people. Which equation represents this situation?

A) $1035x + 535y = 13165$

B) $1035x + 535 = 13165y$

C) $1035x + 535 = 1570y$

D) $1035x + 535y = 1570$

20. [October 2025]

$$5a = \frac{4b^2}{a} - 20a$$

The given equation relates the positive variables a and b. Which equation correctly expresses a in terms of b?

A) $a = \dfrac{4b}{25}$

B) $a = \dfrac{4b^2}{25}$

C) $a = \dfrac{2b}{5}$

D) $a = \dfrac{2b^2}{5}$

21. [October 2025]

Which expression is equivalent to $4x(5xy)$?

A) $20x^2y$

B) $4x^2y$

C) $20x^2y^2$

D) $5x^2y^2$

22. [October 2025]

Which expression is equivalent to $72t^2s^3 - 48t^3s$?

A) $4t^2s(18ts - 12s)$

B) $4t^2s(18s^2 - 12t)$

C) $4ts(18s^2 - 12ts)$

D) $4ts(18t^2s^2 - 12t)$

23. [October 2025]

$$b^2 + 4c = 7d$$

The given equation relates the real numbers b, c, and d, where $d > \frac{4}{7}c$. Which equation correctly expresses b in terms of c and d?

A) $b = \dfrac{7d + 4c}{2}$

B) $b = \dfrac{7d - 4c}{2}$

C) $b = \pm\sqrt{7d + 4c}$

D) $b = \pm\sqrt{7d - 4c}$

24. [October 2025]

$$f(x) = 2x + 3$$
$$g(x) = 7x - 2$$
$$h(x) = 5x + 6$$

The functions f, g, and h are defined as shown. If $f(x) \cdot g(x) - h(x) = ax^2 + bx + c$, where a, b, and c are constants, what is the value of b?

A) -5

B) 12

C) 20

D) 22

Answers: Expressions

Number	Answer	Number	Answer
1	B	2	D
3	D	4	D
5	D	6	D
7	B	8	C
9	C	10	C
11	C	12	A
13	D	14	B
15	A	16	B
17	B	18	C
19	A	20	C
21	A	22	B
23	D	24	B

Linear Equations

1. [August 2025]

If $\frac{x-17}{18} = \frac{x-17}{6}$, what is the value of $x+17$?

A) 0
B) 3
C) 17
D) 34

2. [August 2025]

In the linear function f, $f(0) = 5$ and $f(7) = 5$. Which equation defines f?

A) $f(x) = 0$
B) $f(x) = 7$
C) $f(x) = 5$
D) $f(x) = x+5$

3. [August 2025]

If $4 + \dfrac{5(7-x)}{2} = \dfrac{8(7-x)}{3}$, what is the value of $7-x$?

4. [August 2025]

$$c(x-6) = -7(x+k)$$

In the given equation, c and k are constants. The equation has exactly one solution. Which of the following statements must be true?

A) The value of c cannot be -7.
B) The value of c cannot be $-\frac{7}{6}$
C) The value of k cannot be $\frac{6}{7}$
D) The value of k cannot be -6

5. [August 2025]

Micah deposited a total of \$6,600 into two retirement accounts last year by depositing x dollars once each month into one account and y dollars twice each month into the other account. Which equation represents this situation?

A) $6600 = 12x + 12y$
B) $6600 = 12x + 24y$
C) $6600 = 24x + 12y$
D) $6600 = 24x + 24y$

6. [August 2025]

If $|16 - 2x| = 58$, which of the following is a value of $8 - x$?

A) 0
B) 8
C) 29
D) 42

7. [August 2025]

If $10 + \frac{8(7-x)}{5} = \frac{5(7-x)}{3}$, what is the value of $7-x$?

8. [August 2025]

$$|a+10| = 21$$

What is a possible solution to the given equation?

A) -21
B) -10
C) 0
D) 11

9. [September 2025]

During hibernation, American black bears do not eat or replenish calories. A certain black bear weighed 304 pounds when entering hibernation and lost weight at a rate of 0.6 pound per day.At this rate, how many days after entering hibernation did the black bear weigh 250 pounds?

10. [September 2025]

What is the x-intercept of the graph of $4x + y = 24$ in the xy-plane?

A) $(6, 0)$
B) $(0, 0)$
C) $(28, 0)$
D) $(4, 0)$

11. [September 2025]

$$1.5c + 3d = 42$$

The given equation describes the relationship between the number of cats, c, and the number of dogs, d, that can be cared for at a pet care business during a week. If the business cares for 14 dogs during a week, how many cats can it care for during this week?

A) 0
B) 42
C) 21
D) 3

12. [September 2025]

On January 1, 2000, the population of a town was 26,252 and on January 1, 2010, the population was 26,753. The equation $10x + 26,252 = 26,753$ describes this situation. Which of the following is the best interpretation of x in this context?

A) The percentage by which the population of the town increased each year between 2000 and 2010
B) The total increase in population between 2000 and 2010
C) The projected population of the town 10 years after 2010
D) The average increase per year of the population between 2000 and 2010

13. [September 2025]

$$8(t + 4) - 10(t + 4) = 34$$

What value of t is the solution to the given equation?

14. [September 2025]

$$g = 14 - \frac{x}{23}$$

The equation shown gives the estimated amount of gas g, in gallons, that remains in the gas tank of a car after being driven x miles, where $0 \le x \le 322$. What is the estimated amount of gas, in gallons, that remains in the gas tank of the car when $x = 230$?

A) 14
B) 0
C) 9
D) 4

15. [September 2025]

Number of cars	Maximum number of passengers and crew
2	75
7	245
9	313

The table shows the linear relationship between the number of cars, c, on a commuter train and the maximum number of passengers and crew, p, that the train can carry. Which equation represents the linear relationship between c and p?

A) $34p - c = 7$

B) $34c - p = 7$

C) $34p - c = -7$

D) $34c - p = -7$

16. [September 2025]

If $4(2x - 0.6) + x = 0.45 + 0.3x$, what is the value of $870x$?

17. [September 2025]

$$38x + 85y = 360$$

The given equation represents the volume of mulch x, in cubic feet, and the volume of soil y, in cubic feet, in a mixture of mulch and soil that weighs 360 pounds. If the volume of soil in the mixture is 2 cubic feet, what is the volume of mulch in the mixture, in cubic feet?

18. [September 2025]

A partially filled container containing 28 milliliters of water is placed under a leaky faucet that produces one 0.05-milliliter drop of water every 5 seconds. Until the container is full, which of the following can be used to represent the volume v, in milliliters, of water in the container t seconds after it is placed under the faucet, where t is a multiple of 5?

A) $v = 5t$

B) $v = 0.25t + 28$

C) $v = 0.01t + 28$

D) $v = 0.05t + 28$

19. [September 2025]

$$g = 13 - \frac{x}{29}$$

The equation shown gives the estimated amount of gas g, in gallons, that remains in the gas tank of a car after being driven x miles, where $0 \leq x \leq 377$. What is the estimated amount of gas, in gallons, that remains in the gas tank of the car when $x = 290$?

A) 13

B) 0

C) 3

D) 16

20. [September 2025]

What is the slope of the graph of $y = \frac{7}{12}x$ in the xy-plane?

A) 0

B) $\frac{7}{12}$

C) $-\frac{7}{12}$

D) $\frac{12}{7}$

21. [September 2025]

$$34x + 79y = 260$$

The given equation represents the volume of mulch x, in cubic feet, and the volume of soil y, in cubic feet, in a mixture of mulch and soil that weighs 260 pounds. If the volume of soil in the mixture is 2 cubic feet, what is the volume of mulch in the mixture, in cubic feet?

22. [September 2025]

A partially filled container containing 26 milliliters of water is placed under a leaky faucet that produces one 0.03-milliliter drop of water every 3 seconds. Until the container is full, which of the following can be used to represent the volume v, in milliliters, of water in the container t seconds after it is placed under the faucet, where t is a multiple of 3?

A) $v = 0.01t + 26$

B) $v = 0.09t + 26$

C) $v = 3t$

D) $v = 0.03t + 26$

23. [September 2025]

If $\frac{x+5}{3} = \frac{x+5}{13}$, the value of $x + 5$ is between which of the following pairs of values?

A) 7 and 13

B) 4 and 6

C) -4 and 4

D) -6 and -3

24. [September 2025]

During hibernation, American black bears do not eat or replenish calories. A certain black bear weighed 515 pounds when entering hibernation and lost weight at a rate of 0.5 pound per day. At this rate, how many days after entering hibernation did the black bear weigh 490 pounds?

25. [September 2025]

If $x + 1 = 3$, what is the value of $2(x + 1)$?

26. [September 2025]

$$1.5c + 3d = 108$$

The given equation describes the relationship between the number of cats, c, and the number of dogs, d, that can be cared for at a pet care business during a week. If the business cares for 36 dogs during a week, how many cats can it care for during this week?

A) 3

B) 54

C) 0

D) 108

27. [September 2025]

$$6(t + 7) - 8(t + 7) = 38$$

What value of t is the solution to the given equation?

28. [September 2025]

If $\frac{x+5}{4} = \frac{x+5}{13}$, the value of $x+5$ is between which of the following pairs of values?

A) -2 and 2
B) 7 and 13
C) -6 and -4
D) 2 and 6

29. [October 2025]

$$9x + 17 = 9x + k$$

In the given equation, k is a constant. The equation has infinitely many solutions. What is the value of k?

30. [October 2025]

In the xy-plane, line k passes through the points $(3, 0)$ and $(0, 5)$. Which of the following is an equation of line k?

A) $3x - 5y = 15$
B) $5x + 3y = 15$
C) $3x + 5y = 15$
D) $5x - 3y = 15$

31. [October 2025]

A chemist combines 25 grams of compound A, 35 grams of compound B, and x grams of compound C. The resulting mixture has a mass of 125 grams. Find the value of x.

A) 25
B) 65
C) 35
D) 10

32. [October 2025]

Carmen needs at least 100 signatures to submit a petition. If Carmen already has 51 signatures, what is the minimum number of additional signatures Carmen needs to submit the petition?

A) 41
B) 151
C) 49
D) 51

33. [October 2025]

There are a total of 340 seats in a school auditorium. During an assembly, students occupied 50% of the seats in the auditorium. How many seats did the students occupy during this assembly?

A) 290
B) 25
C) 170
D) 50

34. [October 2025]

A phone plan charges a flat monthly fee plus a fixed rate per minute of calls. In January, a customer made 80 minutes of calls and was billed \$22. In February, the same customer made 200 minutes of calls and was billed \$40. Which of the following represents the monthly bill B, in dollars, as a function of the number of minutes m?

A) $B = 0.10m + 14$
B) $B = 0.15m + 10$
C) $B = 0.12m + 12$
D) $B = 0.20m + 6$

35. [October 2025]

If $5c = 15$, what is the value of $5c - 8$?

A) 3
B) 67
C) 7
D) 23

36. [October 2025]

$$7j = k + 13m$$

The given equation relates the distinct positive numbers j, k, and m. Which equation correctly expresses j in terms of k and m?

A) $j = 7(k + 13m)$
B) $j = \frac{k}{7} + 13m$
C) $j = k + \frac{13m}{7}$
D) $j = \frac{k+13m}{7}$

37. [October 2025]

The function f is defined by $f(x) = \frac{x}{5} + 9$. For what value of x does $f(x) = 13$?

A) 20
B) -32
C) -13
D) -22

38. [October 2025]

How many solutions does the equation $14x = 6x + 8x$ have?

A) Exactly one
B) Zero
C) Infinitely many
D) Exactly two

39. [October 2025]

A designer used a total of 23 yards of blue fabric and yellow fabric for a sewing project. The designer used 17 yards of blue fabric. How many yards of yellow fabric were used?

40. [October 2025]

If $9x + 4 = 67$, what is the value of $90x + 40$?

A) 7
B) 70
C) 130
D) 670

41. [October 2025]

The equation $58 = 2x + 2y$ gives the perimeter of a rectangular garden that has length x, in feet, and width y, in feet. The width of the garden is 14 feet. What is the length, in feet, of the garden?

42. [October 2025]

During hibernation, American black bears do not eat or replenish calories. A certain black bear weighed 293 pounds when entering hibernation and lost weight at a mean rate of 0.9 pounds per day. At this rate, how many days after entering hibernation would the black bear weigh 230 pounds?

A) 57
B) 63
C) 70
D) 207

43. [October 2025]

$$\frac{x}{5} + \frac{y}{9} = \frac{47}{45}$$

An engineer connects resistors in series, where the resistors in the series have a total resistance of $\frac{47}{45}$ ohms. In this series, there are x resistors of type A, which each have a resistance of a ohms, and y resistors of type B, which each have a resistance of b ohms. The given equation represents this situation. According to this equation, what is the positive difference between the value of a and the value of b?

A) 47

B) 4

C) $\frac{47}{45}$

D) $\frac{4}{45}$

44. [October 2025]

$$b(6 - x) = 5x + 12$$

In the equation above, b is a constant. The equation has exactly one solution for x. Which of the following CANNOT be a value of b?

A) -5

B) 0

C) 5

D) 10

45. [October 2025]

The total cost $f(x)$, in dollars, to lease a car for 24 months from a particular car dealership is given by $f(x) = 24x + 1000$, where x is the monthly payment, in dollars. What is the total cost to lease a car when the monthly payment is $300?

A) $6200

B) $7000

C) $8200

D) $25300

46. [October 2025]

$$nx + 7t = -tx + 8n$$

In the given equation, n and t are constants. The equation has no solution. What can be the value of $n + t$?

47. [October 2025]

Each 3.0-ounce serving of cheddar cheese and each 1.2-ounce serving of tuna provides about 1 microgram of vitamin B12. If a total of 1.7 micrograms of vitamin B12 are consumed from eating x ounces of cheese and y ounces of tuna, which equation best represents this situation?

A) $3.0x + 1.2y = 1.7$

B) $0.33x + 0.83y = 1.7$

C) $0.83x + 0.33y = 1.7$

D) $1.2x + 3.0y = 1.7$

48. [August 2025]

If $2x = 8$, what is the value of $10x$?

A) 9

B) 13

C) 20

D) 40

Answers: Linear Equations

Number	Answer	Number	Answer
1	D	2	C
3	24	4	A
5	B	6	C
7	150	8	D
9	90	10	A
11	A	12	D
13	-21	14	D
15	D	16	285
17	5	18	C
19	C	20	B
21	3	22	A
23	C	24	50
25	6	26	C
27	-26	28	A
29	17	30	B
31	B	32	C
33	C	34	B
35	C	36	D
37	A	38	C
39	6	40	D
41	15	42	C

Number	Answer	Number	Answer
43	D	44	A
45	C	46	0
47	B	48	D

Linear System of Equations

1. [August 2025]

$$3x + 6y = 17$$
$$-3x - 4y = 5$$

The solution to the given system of equations is (x, y). What is the value of $2y$?

2. [August 2025]

$$y = -12x + 16$$
$$y = -20x + 24$$

What is the solution (x, y) to the given system of equations?

A) (1,4)
B) (16,24)
C) (24,16)
D) (4,1)

3. [August 2025]

$$2x = 9 + 7y$$
$$-2x = 3 + 7y$$

The solution to the given system of equations is (x, y). What is the value of $4x$?

4. [August 2025]

$$x^2 + y + 3 = 3$$
$$8x + 16 - y = 0$$

The solution to the given system of equations is (x, y). What is the value of x?

A) -16
B) -4
C) 2
D) 8

5. [August 2025]

$$y = 3x + 5,000$$
$$y = -3x + 13,000$$

The solution to the given system of equations is (x, y). What is the value of $2y$?

A) 36000
B) 18000
C) 9000
D) 8000

6. [August 2025]

$$y = (x - 6)(x + 5)$$
$$y = 11x - 66$$

The solution to the given system of equations is (x, y). What is the value of $2y$?

A) 0
B) -5
C) -6
D) 6

7. [August 2025]

$$y = 3x + 11,000$$
$$y = -3x + 13,000$$

The solution to the given system of equations is (x, y). What is the value of 2y?

A) 48000
B) 24,000
C) 12,000
D) 2,000

8. [August 2025]
An artist made 13 pieces of pottery using 42 pounds of clay. The pottery included only bowls and vases. Each bowl used 2 pounds of clay and each vase used 4 pounds of clay. How many vases did the artist make?

A) 5
B) 6
C) 7
D) 8

9. [September 2025]

$$y = -15x + 19$$
$$y = -20x + 24$$

What is the solution (x, y) to the given system of equations?

A) (24, 19)
B) (19, 24)
C) (4, 1)
D) (1, 4)

10. [September 2025]

$$5(9x + 7 + \frac{c}{5}) = 45x$$

In the given equation, c is a constant. The equation has infinitely many solutions. What is the value of c?

11. [September 2025]
If $x + 4y = 31$ and $9x - 20y = -57$, what is the value of y?

12. [September 2025]

$$8x + 32y = 38$$
$$12x + 48y = 57$$

At how many points do the graphs of the given equations intersect in the xy-plane?

A) Exactly one
B) Infinitely many
C) Exactly two
D) Zero

13. [September 2025]
If $x + 3y = 29$ and $7x - 12y = -61$, what is the value of y?

14. [September 2025]

$$8x + 32y = 30$$
$$12x + 48y = 45$$

At how many points do the graphs of the given equations intersect in the xy-plane?

A) zero
B) exactly one
C) infinitely many
D) exactly two

15. [September 2025]

$$y = -10x + 13$$
$$y = -12x + 15$$

What is the solution to the given system of equations?

A) $(13, 15)$
B) $(15, 13)$
C) $(3, 1)$
D) $(1, 3)$

16. [September 2025]

$$3x + y = 21$$
$$9x - y = 3$$

How many solutions does the given system of equations have?

A) Infinitely many
B) Zero
C) Exactly two
D) Exactly one

17. [September 2025]

$$(x + k) + \frac{9}{2}(y + t) + 19 = 0$$
$$(x + k) - \frac{9}{2}(y + t) - 19 = 0$$

In the given system of equations, k and t are constants. The solution to this system is (x, y). What is the value of $18(y + t)$?

A) -76
B) 0
C) 38
D) 9

18. [October 2025]

$$x + 4y = -3$$
$$4x - y = 22$$

The solution to the given system of equations is (x, y). What is the value of $3x - 5y$?

19. [October 2025]

$$0.5x + 1.2y = 90$$
$$1.5x + 3.6y = 27$$

How many solutions does the given system of equations have??

A) Infinitely many
B) Exactly two
C) Exactly one
D) Zero

20. [October 2025]
For a study, a research team measured the heights of 380 king penguins from the Kerguelen Islands and 190 king penguins from the Falkland Islands. The research team determined that the mean height of the 380 king penguins from the Kerguelen Islands was 87 centimeters and the mean height of the 190 king penguins from the Falkland Islands was 93 centimeters. What was the mean height, in centimeters, of all 570 king penguins the research team measured for this study?

A) 91
B) 89
C) 90
D) 88

21. [October 2025]

$$y = \frac{1}{2}x - 7$$
$$\frac{1}{2}y = 3x + 2$$

The solution to the given equations is (x, y). What is the value of $(y - x)$?

22. [October 2025]

$$ax - by = 52$$
$$2ax - 8y = 48$$

In the given system of equations, a and b are constants. The graphs of these equations in the xy-plane intersect at the point $(2, 4)$. What is the value of b?

A) 3
B) 4
C) -8
D) -3

23. [October 2025]

$$x + 6y = 28$$
$$6y = 14$$

The solution to the given system of equations is (x, y). What is the value of x?

24. [October 2025]

$$x^2 + y^2 = 36$$
$$y = mx + \frac{b}{4}$$

In the given system of equations, m and b are negative constants. In the xy-plane, the graphs of the equations in the given system intersect at the point $(-5, y)$, where $y < 0$. Which expression represents the value of b?

A) $-\frac{5m}{4} + \frac{\sqrt{11}}{4}$
B) $\frac{5m}{4} - \frac{\sqrt{11}}{4}$
C) $-20m + 4\sqrt{11}$
D) $20m - 4\sqrt{11}$

25. [October 2025]
A biologist mixed a solution that is 0.3% sodium chloride by mass with a solution that is 0.15% sodium chloride by mass to obtain a new solution, which has a mass of 80 grams and contains 0.21 grams of sodium chloride. How many grams of 0.3% sodium chloride solution did the biologist use?

A) 0.14
B) 20
C) 60
D) 79.86

26. [October 2025]

x	y
0	-10
1	-4
2	2

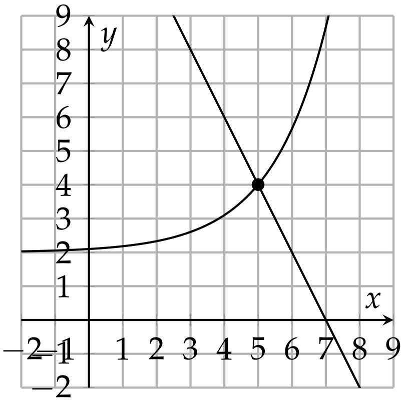
Line k also lies in the xy-plane and is defined by the equation $y = 4x$. At what point (x, y) do lines j and k intersect?

A) (5, 20)
B) (5, -44)
C) (6, 24)
D) (6, 26)

27. [October 2025]
In the given system of equations, t is a constant. If the system has no solution, what is the value of t?

$$4x - 6y = 10y + 2$$
$$ty = \frac{1}{2} + 2x$$

28. [October 2025]



The graph of a system of a linear equation and a nonlinear equation is shown. What is the solution (x, y) to this system?

A) (4,0)
B) (5,4)
C) (0,0)
D) (0,5)

29. [October 2025]

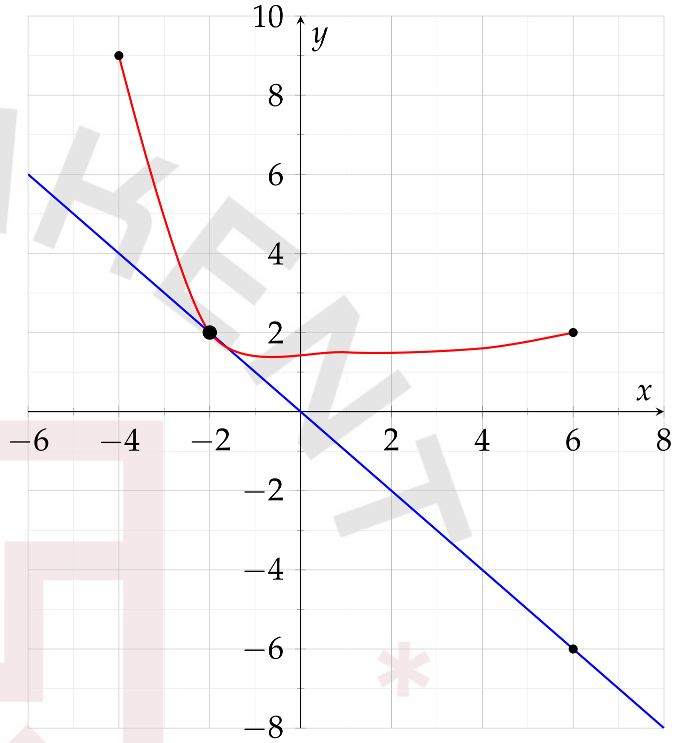
$$0.5x + 1.2y = 9$$
$$1.5x + 3.6y = 27$$

How many solutions does the given system of equations have?

A) Infinitely many
B) Exactly two
C) Exactly one
D) Zero

30. [October 2025]

The graph of a system of a linear equation and a nonlinear equation is shown. What is the solution (x, y) to this system?



A) $(0, 0)$
B) $(0, -2)$
C) $(-2, 2)$
D) $(3, 0)$

31. [September 2025]

$$9x + y = 22$$
$$10x - y = 16$$

How many solutions does the given equations have?

A) Exactly one
B) Infinitely many
C) Zero
D) Exactly two

32. [September 2025]

$$x + 5 = 14$$
$$y = 3x^2 + 3$$

At what point (x, y) do the graphs of the equations in the given system intersect?

A) $(9, 243)$
B) $(14, 591)$
C) $(9, 246)$
D) $(14, 3)$

Answers: Linear System of Equations

Number	Answer	Number	Answer
1	22	2	A
3	6	4	B
5	B	6	A
7	B	8	D
9	D	10	-35
11	6	12	B
13	8	14	C
15	D	16	D
17	A	18	25
19	D	20	B
21	-6	22	D
23	14	24	D
25	C	26	A
27	8	28	B
29	A	30	C
31	A	32	C

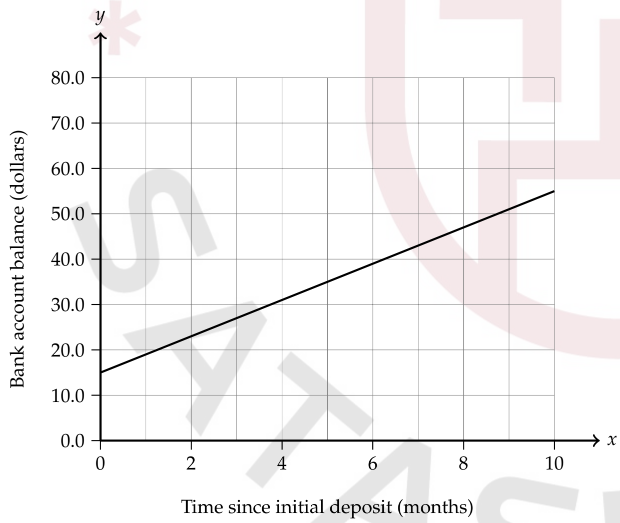
Linear Functions

1. [August 2025]

The function $f(x) = -30x + 310$ gives the predicted height above the ground $f(x)$, in feet,of a model airplane a minutes after it begins to descend. What is the predicted height above the ground, in feet, of the model airplane 2 minutes after it begins to descend?

A) 60
B) 250
C) 280
D) 370

2. [August 2025]



A bank account was opened with an initial deposit. Over the next several months, regular deposits were made into this account, and there were no withdrawals made during this time. The graph of the function f shown, where $y = f(x)$, estimates the account balance, in dollars, in this bank account x months since the initial deposit. To the nearest whole dollar, what is the amount of the initial deposit estimated by the graph?

3. [August 2025]

In the xy-plane, line s passes through the point $(0, 0)$ and is parallel to the line represented by the equation $y = 36x + 6$. If line s also passes through the point $(9, d)$, what is the value of d?

4. [August 2025]

The function f is defined by $f(x) = 25x + 70$. What is the value of $f(x)$ when $x = 2$

A) 50
B) 97
C) 120
D) 190

5. [August 2025]

The table below gives the number of hours, h, of labor and a plumber's total charge $f(h)$, in dollars, for two different jobs.

Hours (h)	Total Charge $f(h)$
1	120
3	220

There is a linear relationship between h and $f(h)$. Which equation represents this relationship?

A) $f(h) = 20h + 100$
B) $f(h) = 100h + 20$
C) $f(h) = 50h + 70$
D) $f(h) = 70h + 50$

6. [August 2025]

What is the slope of the graph of $60x - 10y = -48$ in the xy-plane?

A) -6

B) $-\frac{5}{4}$

C) $\frac{5}{4}$

D) 6

7. [August 2025]

In the relationship between variables x and y, each increase of 8 in the value of x decreases the value of y by 3. When the value of x is 16, the value of y is 10. Which equation represents this relationship?

A) $y = -\frac{3}{8}x + 10$

B) $y = -\frac{3}{8}x + 16$

C) $y = -\frac{8}{3}x + 10$

D) $y = -\frac{8}{3}x + \frac{158}{3}$

8. [August 2025]

Javier deposits \$45 in a savings account at the end of each week. At the beginning of the 1st week of a year there was \$900 in that savings account. How much money, in dollars, will be in the account at the end of the 4th week of the year?

A) 720

B) 945

C) 949

D) 1,080

9. [August 2025]

Line k contains the points $(-3, -48)$, $(v, 0)$, and $(5, 56)$. What is the value of v?

10. [August 2025]

Kai used fabric measuring 4 yards in length to make each costume for a school play. The relationship between the number of costumes that Kai made, x, and the total length of fabric that he purchased y, in yards, is represented by the equation $y - 4x = 5$. What is the best interpretation of 5 in this context?

A) Kai made 5 costumes

B) Kai purchased a total of 5 yards of fabric.

C) Kai used a a total of 5 yards of fabric to make the costumes.

D) Kai purchased 5 yards more fabric than he used to costumes.

11. [August 2025]

During the first 8.00 seconds after a car started moving, its speed increased to 11.0 meters per second. From 8.00 seconds to 14.0 seconds after the car started moving, its speed increased from 11.0 meters per second to 23.0 meters per second. To the nearest hundredth, what is the positive difference between the average rate of change of the speed of the car, in meters per second per second, during the first 8.00 seconds after the car started moving and the average rate of change of the speed of the car, in meters per second per second, from 8.00 seconds to 14.0 seconds after the car started moving?

12. [August 2025]

The function m is defined by $m(x) = 3x + 6$ and the function p is defined by $p(x) = 6 - x$ What is the value of $2m(6) - p(6)$?

13. [August 2025]

x	y
0	n
5	$n+24$
10	$n+48$

There is a linear relationship between x and y. The table shows three values of x and their corresponding values of y in terms of a constant n. What is the slope of the line that represents this relationship in the xy-plane?

14. [August 2025]

A book-of-the-month club charges $16 for the first book purchased during the month and an additional $11 for each book after the first. Which equation describes this situation, where y is the total cost, in dollars, for books purchased during a month, x is the number of books purchased during the month, and $x > 0$?

A) $y = x + 16$

B) $y = 11x + 16$

C) $y = 11(x-1) + 16$

D) $y = (x-1) + 16$

15. [August 2025]

What is the slope of the graph of $48x - 6y = -32$ in the $xy - plane$?

A) -8

B) $-\frac{3}{2}$

C) $\frac{3}{2}$

D) 8

16. [August 2025]

The table gives the number of hours, h, of labor and a plumber's total charge $f(h)$, in dollars, for two different jobs.

h	$f(h)$
1	125
3	225

There is a linear relationship between h and $f(h)$. Which equation represents this relationship?

A) $f(h) = 25h + 100$

B) $f(h) = 100h + 25$

C) $f(h) = 50h + 75$

D) $f(h) = 75h + 50$

17. [August 2025]

The function f is defined by $f(x) = 7x - 65$. If the point $(k, 2k)$ lies on the graph of $y = f(x)$ in the xy-plane, where k is a constant, what is the value of k?

A) 7

B) 14

C) 13

D) 165

18. [October 2025]

Charles saves $\frac{3}{5}$ of the $225 he earns each week from his summer job. If Charles continues to save at this rate, how much money, in dollars, will Charles save in 3 weeks?

19. [August 2025]

In the relationship between variables x and y, each increase of 7 in the value of x decreases the value of y by 3. When the value of x is 14, the value of y is 9. Which equation represents this relationship?

A) $y = \frac{-3}{7}x + 9$

B) $y = \frac{-3}{7}x + 15$

C) $y = \frac{-7}{3}x + 9$

D) $y = \frac{-7}{3}x + \frac{125}{3}$

20. [August 2025]

x	g(x)
1	44
2	38
3	32
4	26

For the linear function g, the table shows four values of x and their corresponding values of $g(x)$. The function can be written as $g(x) = mx + b$, where m and b are constants. What is the value of b?

A) 6

B) 22

C) 44

D) 50

21. [September 2025]

For the linear function f, the graph of $y = f(x)$ in the xy-plane passes through the points (0,6) and (7, 7). What is the slope of $y = f(x)$?

22. [September 2025]

There is a linear relationship between the mass of an object attached to a vertical spring and the length of the spring extension. The table shows data collected after certain objects were each attached to the spring.

Mass of object	Length of spring extension
20 grams	7 cm
60 grams	21 cm

What is the length of the spring extension, in centimeters, when a 50 gram object is attached to the spring?

A) 24.5

B) 34

C) 14

D) 17.5

23. [August 2025]

Line p is defined by $kx + 8y = 17$. Line p passes through the points $(r, 0)$ and $(0, t)$ in the xy-plane, where k, r, and t are nonzero constants and r is not equal to t. Which expression represents the value of k?

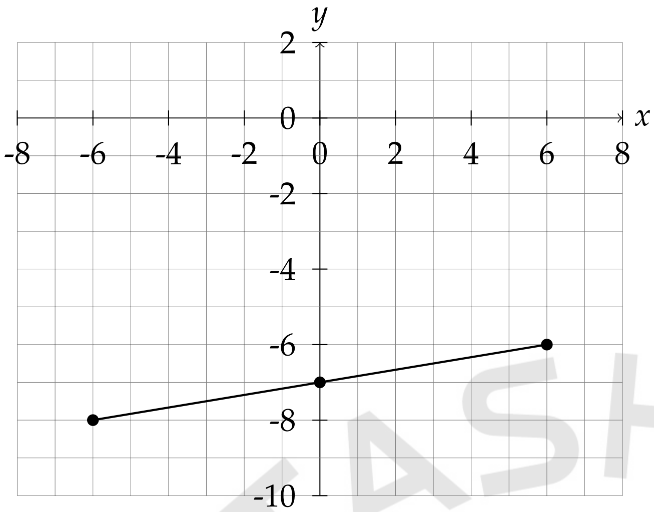
A) $\frac{17}{t}$

B) $\frac{17}{r}$

C) $\frac{8}{t}$

D) $\frac{8}{r}$

24. [October 2025]



- The line slants gradually up from left to right.
- The line passes through the following points:
  - (-6,-8)
  - (0,-7)
  - (6,-6)

The graph of the linear function f is shown, where $y = f(x)$. What is the slope of the graph of f?

A) $\frac{1}{6}$

B) -7

C) -$\frac{1}{6}$

D) 7

25. [October 2025]

$$\frac{q}{t} = \frac{9q}{s} + 5$$

In the given equation, q, r, s, and t are positive, and $\frac{q}{t}$ is greater than 5. Which expression is equivalent to s?

A) $\dfrac{-9qt}{q-5t}$

B) $\dfrac{9qt}{q-5t}$

C) $18qrt$

D) $\dfrac{2qt}{q-5t}$

26. [October 2025]

For certain air temperatures, the table gives wind chill temperatures for two different wind speeds.

Air temperature	Wind chill temperature at wind speed 5 mph	Wind chill temperature at wind speed 15 mph
27 °F	21 °F	15 °F
29 °F	24 °F	18 °F
31 °F	26 °F	20 °F

According to the table, what is the wind chill temperature, in degrees Fahrenheit (°F), when the air temperature is 26 Fahrenheit (°F) and the wind speed is 15 mph? (Disregard the degree when entering your answer) ?

27. [October 2025]

A city has an area of x square miles and the population density of 1035 people per square mile. The suburb adjacent to the city has an area of y square miles and the population density of 535 people per square mile. The combined population of the city and the suburb adjacent to the city is 13,165 people. Which equation represents this situation?

A) $1035x + 535y = 13165$

B) $1035x + 535 = 13165y$

C) $1035x + 535 = 1570y$

D) $1035x + 535y = 1570$

28. [October 2025]

The function $f(x) = -40x + 340$ gives the predicted height above the ground $f(x)$, in feet, of a model airplane a minutes after it begins to descend. What is the predicted height above the ground, in feet, of the model airplane 3 minutes after it begins to descend?

A) 120

B) 220

C) 300

D) 460

29. [August 2025]

A company developed a plan to set the selling price of a product. The company determined that for a selling price of \$120.00, zero products would be sold. For each \$2.50 decrease in the selling price, the number of products sold would increase by one. For a revenue of exactly \$1,437.50, which of the following could be the number of products sold? (revenue = price x number of products sold)

A) 23

B) 24

C) 527

D) 1440

30. [October 2025]

In the xy-plane, what is the slope of the line that passes through the points (0,0) and (13,14)?

A) 13

B) $\frac{14}{13}$

C) $\frac{13}{14}$

D) 14

31. [October 2025]

x	y
$-2s$	17
$-s$	14
s	8

The table shows three values of a and their corresponding values of y, where s is a constant. There is a linear relationship between a and y. Which of the following equations represents this relationship?

A) $sx + 3y = 11s$

B) $3x + sy = 11s$

C) 5

D) 10

32. [September 2025]

Number of cars	Maximum number of passengers and crew
2	102
6	294
10	486

The table shows the linear relationship between the number of cars, c, on a commuter train and the maximum number of passengers and crew, p, that the train can carry. Which equation represents the linear relationship between c and p?

A) $48p - c = -6$

B) $48c - p = 6$

C) $48p - c = 6$

D) $48c - p = -6$

Answers: Linear Functions

Number	Answer	Number	Answer
1	B	2	15
3	324	4	C
5	C	6	D
7	B	8	D
9	0.692; 9/13	10	D
11	0.63	12	48
13	24/5; 4.8	14	C
15	D	16	C
17	C	18	405
19	B	20	B
21	1/7	22	D
23	B	24	A
25	B	26	18
27	A	28	B
29	A	30	B
31	**B**	32	D

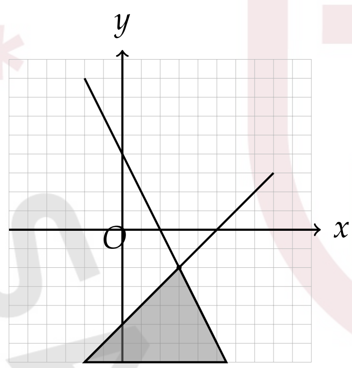
Linear Inequalities

1. [August 2025]

In a set of three consecutive integers, where the integers are ordered from least to greatest, the first integer is represented by x. The sum of 4 and the second integer is less than the product of 17 and the third integer. Which inequality represents this situation?

A) $4 + (x + 1) > 17(x + 2)$
B) $4 + (x + 1) < 17(x + 2)$
C) $4 + (x + 2) > 17(x + 3)$
D) $4 + (x + 2) < 17(x + 3)$

2. [August 2025]



The graphs of $y = -3x + 7$ and $y = x - 5$ are shown. Point P (not shown) has coordinates (-3, 5) and lies in the shaded region. The coordinates of P satisfy which of the following inequality?

I. $y < -3x + 7$
II. $y > x - 5$

A) I only
B) II only
C) I and II
D) Neither I nor II

3. [August 2025]

A number x is less than 8 more than $\frac{1}{2}$ times the value of a number y. If y is an integer and $x = 17$, what is the least possible value of y?

4. [August 2025]

To win a game show, a contestant needs to score at least 70 total points from two rounds. Correct responses in the first round are worth 4 points each, and correct responses in the second round are worth 5 points each. Which inequality models this situation, where f is the number of correct responses in the first round and s is the number of correct responses in the second round?

A) $f + s \leq 70$
B) $4f + 5s \geq 70$
C) $5f + s \geq 70$
D) $5f + 4s \leq 70$

5. [September 2025]

A freight elevator can hold a maximum weight of 5,340 pounds during one trip. A 190-pound person needs to deliver several boxes using the freight elevator. Some of these boxes weigh 25 pounds each and the others weigh 64 pounds each. Which inequality represents the possible combinations of the number of 25-pound boxes, x, and the number of 64-pound boxes, y, the person can deliver during one trip if only the person and the boxes are on the freight elevator?

A) $64x + 25y \geq 5340$
B) $64x + 25y \leq 5340$
C) $25x + 64y \geq 5150$
D) $25x + 64y \leq 5150$

6. [September 2025]

$$y > 2x - 90$$
$$y < -\frac{1}{6}x - 8$$

For which of the following tables are all the values of x and their corresponding values of y solutions to the given system of inequalities?

A)

x	y
18	-8
24	-9
30	-10

B)

x	y
18	-14
24	-15
30	-16

C)

x	y
18	-11
24	-12
30	-13

D)

x	y
18	-57
24	-58
30	-59

7. [September 2025]

$$y > \frac{9}{2}x + b$$

In the given inequality, b is a positive constant. Which of the following does NOT contain any points (x, y) in the xy-plane that are solutions to this inequality?

A) The region where $x > 0$ and $y > 0$
B) The region where $x < 0$ and $y < 0$
C) The region where $x < 0$ and $y > 0$
D) The region where $x > 0$ and $y < 0$

8. [September 2025]

$$y > \frac{9}{7}x + b$$

In the given inequality, b is a positive constant. Which of the following does NOT contain any points (x, y) in the xy-plane that are solutions to this inequality?

A) The region where $x < 0$ and $y < 0$
B) The region where $x < 0$ and $y > 0$
C) The region where $x > 0$ and $y > 0$
D) The region where $x > 0$ and $y < 0$

9. [October 2025]

In the xy-plane, which of the following does NOT contain any points that are part of the solution set to $5x - 7y > 35$?

A) The x-axis
B) The region where $x > 0$ and $y > 0$
C) The region where $x < 0$ and $y < 0$
D) The region where $x < 0$ and $y > 0$

10. [September 2025]

$$y < x$$
$$y > -5x - 10$$

For which of the following tables are all the values of x and their corresponding values of y solutions to the given system of inequalities?

A)

x	y
6	11
7	10
8	9

B)

x	y
6	4
7	5
8	6

C)

x	y
-13	-15
-12	-14
-11	-13

D)

x	y
-13	-6
-12	-7
-11	-8

11. [September 2025]

$$y < x$$
$$y > -4x - 9$$

For which of the following tables are all the values of x and their corresponding values of y solutions to the given system of inequalities?

A)

x	y
5	4
6	5
7	6

B)

x	y
-12	-5
-11	-6
-10	-7

C)

x	y
-12	-13
-11	-12
-10	-11

D)

x	y
5	10
6	9
7	8

12. [September 2025]

A freight elevator can hold a maximum weight of 5,450 pounds during one trip. A 190-pound person needs to deliver several boxes using the freight elevator. Some of these boxes weigh 23 pounds each and the others weigh 60 pounds each. Which inequality represents the possible combinations of the number of 23-pound boxes, x, and the number of 60-pound boxes, y, the person can deliver during one trip if only the person and the boxes are on the freight elevator?

A) $60x + 23y \geq 5450$
B) $23x + 60y \geq 5260$
C) $60x + 23y \leq 5450$
D) $23x + 60y \leq 5260$

13. [September 2025]

$$y > 2x - 72$$

$$y < \frac{-1}{6}x - 9$$

For which of the following tables are all the values of x and their corresponding values of y solutions to the given system of inequalities?

A)

x	y
12	-11
18	-12
24	-13

B)

x	y
12	-14
18	-15
24	-16

C)

x	y
12	-57
18	-58
24	-59

D)

x	y
12	-8
18	-9
24	-10

14. [October 2025]

A pet service that takes care of both cats and dogs can care for at most 43 animals. Which inequality represents this situation, where c is the number of cats and d is the number of dogs?

A) $c - d \leq 43$
B) $c - d \geq 43$
C) $c + d \geq 43$
D) $c + d \leq 43$

15. [October 2025]

Carmen needs at least 100 signatures to submit a petition. If Carmen already has 51 signatures, what is the minimum number of additional signatures Carmen needs to submit the petition?

A) 41
B) 151
C) 49
D) 51

16. [October 2025]

An environmental scientist needs to collect at least 38 samples of red clay soil for analysis. If r represents the number of samples the scientist collects for analysis, which inequality represents this situation?

A) $r \leq 38$
B) $r - 38 \geq 38$
C) $r - 38 \leq 38$
D) $r \geq 38$

17. [October 2025]

An agricultural scientist needs to collect at least 200 soil samples in a certain area for a study on soil quality. If the scientist has already collected 168 soil samples, what is the minimum number of additional soil samples the scientist needs to collect for the study?

A) 368
B) 12
C) 148
D) 32

18. [October 2025]

Zuri has a goal to run at least 16 miles per week while training for a race. This week, she has run 4 miles. If x represents the additional number of miles Zuri needs to run this week to meet her goal, which inequality represents this situation?

A) $4 - x \leq 16$
B) $4 + x \leq 16$
C) $4 - x \geq 16$
D) $4 + x \geq 16$

Answers: Linear Inequalities

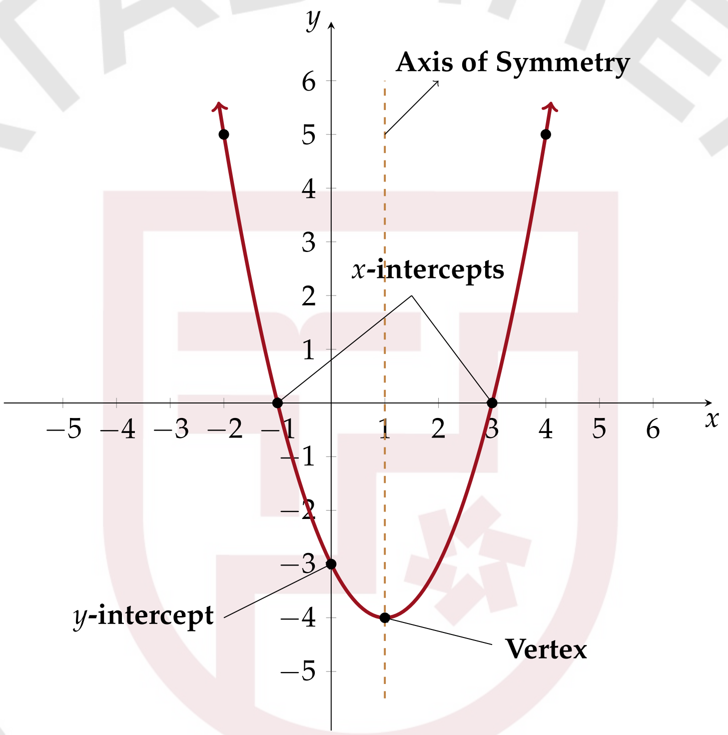
Number	Answer	Number	Answer
1	B	2	C
3	19	4	B
5	D	6	B
7	D	8	D
9	D	10	B
11	A	12	D
13	B	14	D
15	c	16	D
17	D	18	D

Advanced math

Advanced Math Formulas

Quadratic Functions-Forms

- **Standard Form:** $y = ax^2 + bx + c$, where:
  - $c - y$-intercept.
- **Factored Form:** $y = a(x - x_1)(x - x_2)$, where:
  - $x_1, x_2 -$ x-intercepts.
- **Vertex Form:** $y = a(x - h)^2 + k$, where:
  - (h, k)-vertex coordinate.

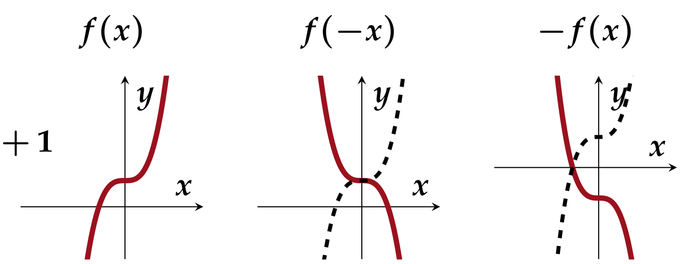


Axis of symmetry and x-coordinate of vertex: $-\dfrac{b}{2a}$

Graph Reflection Rules

- $-f(x) \to$ Reflection about the x-**axis**
- $f(-x) \to$ Reflection about the y-**axis**

When $f(x) = x^3 + 1$



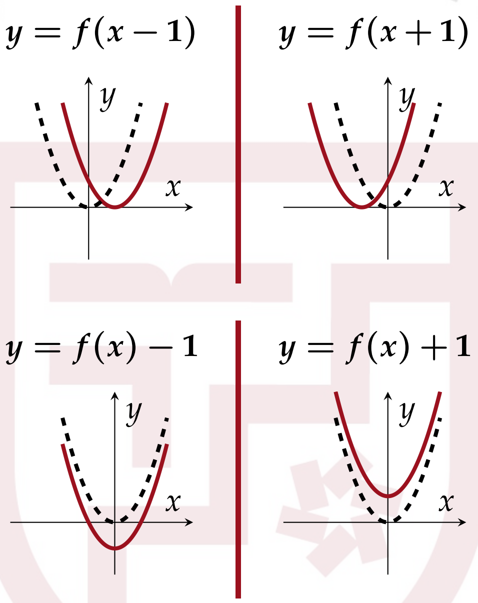
- **Midpoint** of $A(x_1, y_1), B(x_2, y_2)$:

$$\left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2}\right).$$

- **Distance formula:** $\sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$.

Function Translation Rules

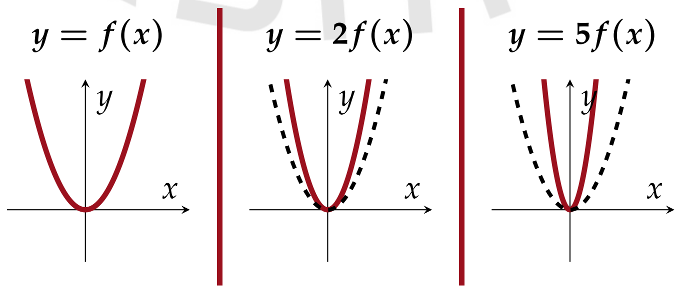
- $f(x+1)$ graph moves **LEFT** one.
- $f(x-1)$ graph moves **RIGHT** one.
- $f(x)+1$ graph moves **UP** one.
- $f(x)-1$ graph moves **DOWN** one.



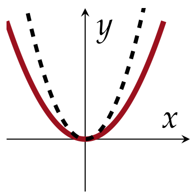
Graph Amplification Rules

x**-intercepts remain the same, but for other points:**

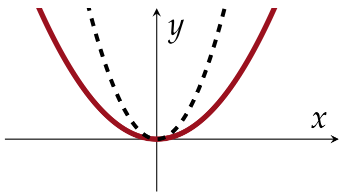
- $2f(x)$ graph moves **farther** from x-axis.
- $\frac{1}{2}f(x)$ graph moves **closer** to x-axis.



$y = \frac{1}{2}f(x)$



$y = \frac{1}{3}f(x)$



Polynomials

1. [September 2025]

For the polynomial function f, the grapf of $y = f(x)$ in the xy-plane passes through the points (-3,0), (1,0) and (4,0).Which of the following must be a factor of $f(x)$?

A) $x+1$

B) $x+4$

C) $x-1$

D) $x-3$

2. [August 2025]

$$x(x-14)(x+9)(x-5)^2 = 0$$

What is the greatest solution to the given equation?

3. [August 2025]

$$x(x-13)(x+9)(x-5)^2 = 0$$

What is the greatest solution to the given equation?

4. [August 2025]

Which expression is a factor of $x^5 + x^4y + xy^8 + y^9$?

A) $x^4 + y^8$

B) $x^4 + y^9$

C) $x^5 + y^8$

D) $x^5 + y^9$

5. [September 2025]

Which equation is equivalent to $v^4 - 450v^3$?

A) $(v^2 - 45)(v^2 - 10)$

B) $(v^2 + 45)(v^2 - 10)$

C) $v^3(v + 450)$

D) $v^3(v - 450)$

6. [September 2025]

For the polynomial function f, the graph of $y = f(x)$ in the xy-plane passes through the points (-4,0), (3,0), and (5,0). Which of the following must be a factor of $f(x)$?

A) $x+5$

B) $x+3$

C) $x-3$

D) $x-4$

7. [September 2025]

For the polynomial function f, the graph of $y = f(x)$ in the xy-plane passes through the points (-4, 0), (1, 0), and (9, 0). Which of the following must be a factor of $f(x)$?

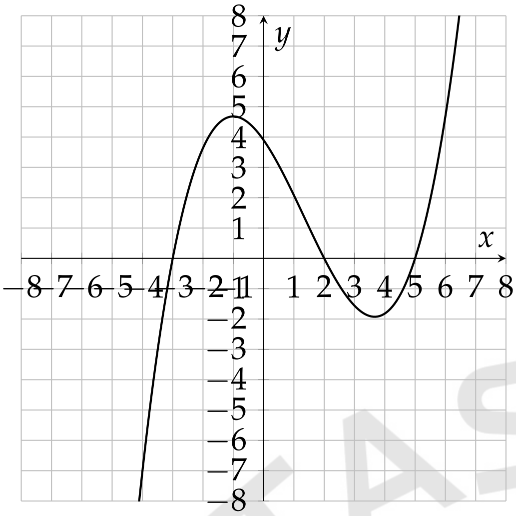
A) $x-1$

B) $x+9$

C) $x-4$

D) $x+1$

8. [September 2025]



The graph of the polynomial function f is shown, where $y = f(x)$. What is the value of $f(5)$?

A) 2

B) 5

C) -3

D) 0

9. [October 2025]

$$-4(v-9)(v-2)(v+7) = 0$$

What value of v is a solution to the given equation?

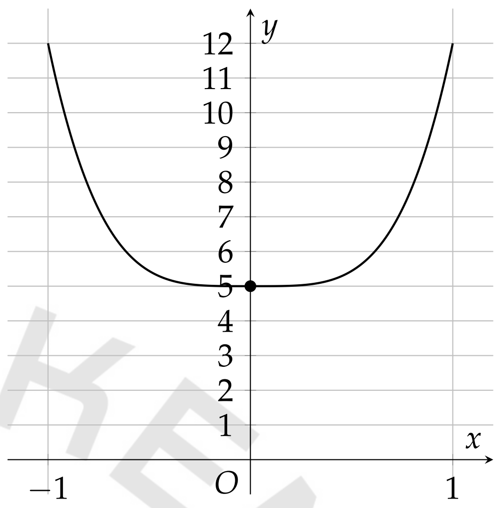
A) -2

B) 7

C) 9

D) -4

10. [October 2025]



The graph of the polynomial function f, where $y = f(x)$, is shown. The y-intercept of the graph is $(0, y)$. What is the value of y?

11. [October 2025]

Which expression is a factor of $3x^2 + 57x + 15$?

A) $3x^2$

B) $5x$

C) 3

D) $57x$

12. [October 2025]

$$5a = \frac{4b^2}{a} - 20a$$

The given equation relates the positive variables a and b. Which equation correctly expresses a in terms of b?

A) $a = \frac{4b}{25}$

B) $a = \frac{4b^2}{25}$

C) $a = \frac{2b}{5}$

D) $a = \frac{2b^2}{5}$

13. [October 2025]

$$\frac{x}{(x-1)(x+1)} = \frac{25}{16x}$$

Which value is a solution to the given equation?

A) $\frac{5}{3}$

B) $\frac{1}{6}$

C) $\frac{5}{18}$

D) 1

14. [October 2025]

$$9x + x = 7x + x + 9$$

Which of the following equations has the same solution as the given equation?

A) $16x = -9$

B) $16x = 9$

C) $2x = 9$

D) $2x - 9$

Answers: Polynomials

Number	Answer	Number	Answer
1	C	2	14
3	13	4	A
5	D	6	C
7	A	8	D
9	C	10	5
11	C	12	C
13	A	14	C

Exponents&Radicals

1. [September 2025]

$$\sqrt{w} + 22 = 28$$

What is the solution to the given equation?

A) 2
B) 3
C) 12
D) 36

2. [September 2025]

For what value of a is $\sqrt[a]{r^{42}}$ equivalent to $r^{6/7}$, where $r > 1$.

3. [August 2025]

$$53 = \frac{2K}{v^2}$$

For an object with a mass of 53 kilograms, the given equation relates the kinetic energy K, in joules, of the object to the object's speed u, in meters per second, where K and v are positive. Which equation correctly expresses this object's speed, in meters per second, in terms of the object's kinetic energy, in joules?

A) $v = \sqrt{\frac{2K}{53}}$
B) $v = \sqrt{\frac{53}{2k}}$
C) $v = \frac{53^2}{2K}$
D) $v = \frac{2K}{53^2}$

4. [September 2025]

The expression $\frac{\sqrt[7]{p^5}}{\sqrt{p^{t+3}}}$, where t is a constant, is equivalent to $\sqrt[7]{p^2}$ for all positive values of p. What is the value of t?

A) $-\frac{15}{7}$
B) $-\frac{27}{7}$
C) $-\frac{24}{7}$
D) $-\frac{18}{7}$

5. [September 2025]

$$\sqrt{x^2} = 24 - 5x$$

A) 2
B) 4
C) 19
D) 30

6. [September 2025]

Which expression is equivalent to $\frac{1}{5y^8}$, where $y > 1$?

A) $\frac{y^{-8}}{5}$
B) $(5y)^{-8}$
C) $5y^{-8}$
D) $\frac{5}{y^{-8}}$

7. [September 2025]

$$\sqrt{x^2} = 80 - 9x$$

What is the solution to the given equation?

A) 4
B) 71
C) 90
D) 8

Answers: Exponents&Radicals

Number	Answer	Number	Answer
1	D	2	49
3	A	4	A
5	B	6	A
7	D		

Functions&Function Notation

1. [September 2025]

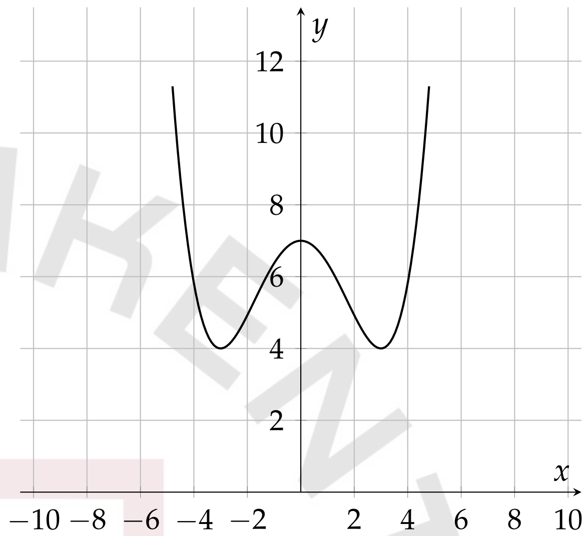
The function f is defined by $f(x) = 7x^3 + 6$. What is the value of $f(2)$?

2. [September 2025]

The function g is defined by $g(x) = 19 - x$. What is the value of $g(4)$?

A) 4
B) 15
C) 19
D) 23

3. [August 2025]



- Moving from left to right, the curve passes from quadrant 2 to quadrant 1
- The curve has 2 relative minimums and 1 relative maximum.
- The curve passes through the following points:
- (negative 3 comma 4)
- (0 comma 7)
- (3 comma 4)

The graph of the function f is shown, where $y = f(x)$. What is the value of f(0)?

A) -7
B) -3
C) 3
D) 7

4. [August 2025]

The function f is defined by $f(x) = 25x + 20$. What is the value of $f(x)$ when $x = 2$?

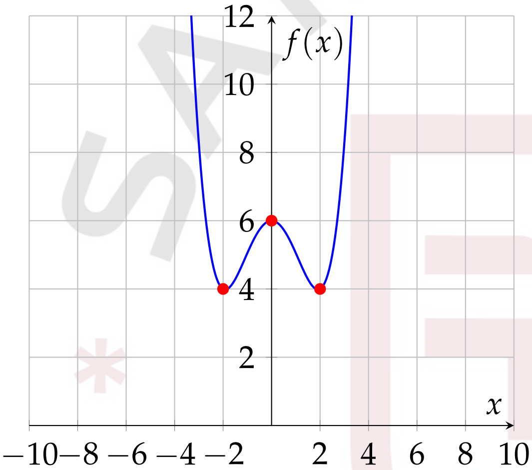
A) 47

B) 50

C) 70

D) 90

5. [August 2025]



The graph of the function f is shown, where $y = f()$. What is the value of $f(0)$?

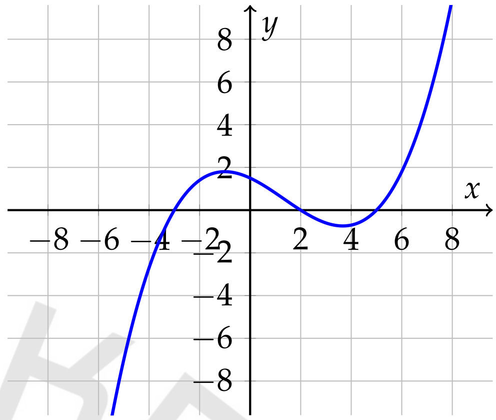
A) 6

B) 2

C) −2

D) −6

6. [September 2025]



The graph of the polynomial function f is shown, where $y = f(x)$. What is the value of $f(2)$?

A) 2

B) 0

C) 5

D) −3

7. [September 2025]

$$f(x) = 4(g(x)) - 2$$

$$g(x) = |13x - 8|$$

The functions f and g are defined by the given equations. What is the value of $f(-10)$?

A) −554

B) −42

C) 550

D) 138

8. [September 2025]

The function f is defined by $f(x) = 20x^3$. The graph of $y = f(-x) + c$ in the xy-plane, where c is a positive integer constant, has an x-intercept at $(r, 0)$ and a y-intercept at $(0, t)$, where r and t are constants. Which of the following must be true about r and t?

A) $r > 0$ and $t > 0$
B) $r > 0$ and $t < 0$
C) $r < 0$ and $t > 0$
D) $r < 0$ and $t < 0$

9. [September 2025]

$$f(x) = 5(g(x)) - 3$$
$$g(x) = |13x - 9|$$

The functions f and g are defined by the given equations. What is the value of $f(-10)$?

A) 692
B) 139
C) 698
D) -53

10. [September 2025]

The function f is defined by $f(x) = 11x^3$. The graph of $y = f(-x) + c$ in the xy-plane, where is c a positive integer constant, has an x-intercept at (r, 0) and a y-intercept at (0, t), where and are constants. Which of the following must be true about r and t?

A) $r < 0$ and $t < 0$
B) $r < 0$ and $t > 0$
C) $r > 0$ and $t > 0$
D) $r > 0$ and $t < 0$

11. [September 2025]

The function w is defined by $w(r) = \frac{1}{r-8} - \frac{r-5}{-r+4.25}$. What is the greatest possible value of r such that $w(r) = 0$?

12. [September 2025]

$$f(x) = 3d(26x + 27) + 18$$

Which of the following represents the x-intercept of the graph of $y = f(x) + 6$ in the xy-plane, where d is a constant?

A) $(81d + 24, 0)$
B) $(\frac{-81d-24}{78d}, 0)$
C) $(6 - \frac{81d+18}{78d}, 0)$
D) $(\frac{-45}{78d+6}, 0)$

13. [October 2025]

The function h is defined by $h(x) = 5(9x - 10)$. What is the value of $h(10)$?

A) 400
B) 10
C) 50
D) 90

14. [October 2025]

The function $A(b) = 7b^2$ gives the area of a right triangle, in square centimeters (cm^2), if the length of its base is b cm and the length of its height is 14 times the length of its base. Which of the following is the best interpretation of $A(18) = 2{,}268$?

A) If the length of the base of the triangle is 2,268 cm, then the length of the height of the triangle is 18 cm.

B) If the length of the base of the triangle is 18 cm, then the length of the height of the triangle is 2,268 cm.

C) If the length of the base of the triangle is 18 cm, then the area of the triangle is 2,268 cm^2.

D) If the length of the base of the triangle is 2,268 cm, then the area of the triangle is 18 cm^2.

15. [October 2025]

The function $f(x) = -40x + 340$ gives the predicted height above the ground $f(x)$, in feet, of a model airplane a minutes after it begins to descend. What is the predicted height above the ground, in feet, of the model airplane 3 minutes after it begins to descend?

A) 120

B) 220

C) 300

D) 460

16. [October 2025]

The function f is defined by $f(x) = 2x^3 + \frac{1}{3}x^2 - x$. What is the value of $f(1)$?

A) $\frac{4}{3}$

B) 2

C) $\frac{10}{3}$

D) 6

17. [October 2025]

Function f is defined as $f(x) = (2x - 5)^2$, and function g is defined as $g(x) = 2(2x + 7)$. Which of the following is equivalent to $f(x) + g(x)$?

A) $4x^2 - 16x + 11$

B) $4x^2 - 16x + 39$

C) $4x^2 - 6x + 39$

D) $4x^2 + 24x + 11$

18. [October 2025]

Harper deposited a fixed amount into her bank account each month. The function $f(t) = 400 + 30t$ gives the amount, in dollars, in Harper's bank account after t monthly deposits. What is the best interpretation of 30 in this context?

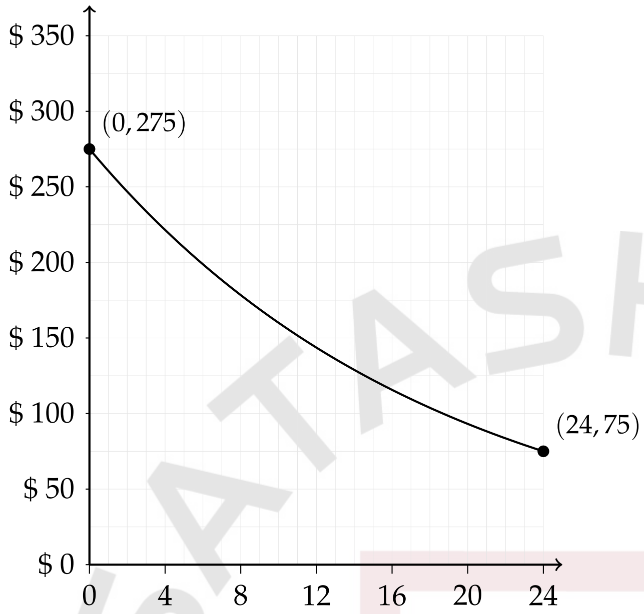
A) With each monthly deposit, the amount in Harper's bank account increased by \$30.

B) Before Harper made any monthly deposits, the amount in her bank account was \$30.

C) After one monthly deposit, the amount in Harper's bank account was \$30.

D) Harper made a total of 30 monthly deposits.

19. [October 2025]



The graph shown gives the estimated value, in dollars, of a mountain bike as a function of the number of months since it was purchased. What is the best interpretation of the y-intercept of the graph in this context?

A) The estimated value of the bike decreased by approximately 2.75% each year after it was purchased.

B) The estimated value of the bike had decreased by $275 in the 24 months after it was purchased.

C) The estimated value of the bike 24 months after it was purchased was $275.

D) The estimated value of the bike was $275 when it was purchased.

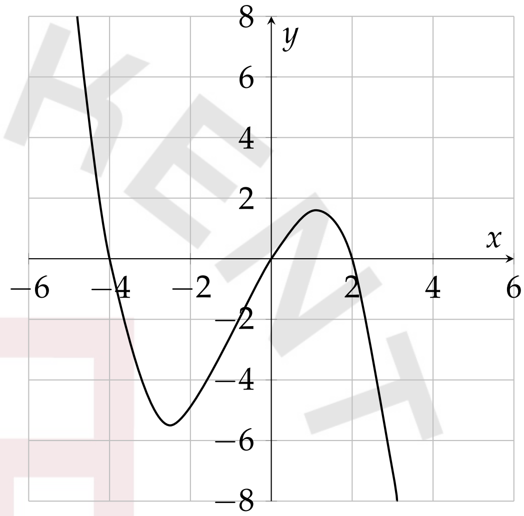
20. [October 2025]

The function f is defined by $f(x) = x^3 + 7$. What is the value of $f(2)$?

21. [October 2025]

The function f is defined by $f(x) = \dfrac{1}{8x}$. What is the value of $f(x)$ when $x = 9$?

22. [October 2025]



The complete graph of $y = f(x)$ is shown. For how many values of x does $f(x) = 0$?

A) One

B) Two

C) Three

D) Four

23. [October 2025]

The function is defined by $f(x) = \dfrac{x+11}{5}$, and $f(a) = -18$, where a is a constant. What is the value of a?

A) -101

B) -79

C) $-\frac{79}{5}$

D) $-\frac{7}{5}$

Answers: Functions&Function Notation

Number	Answer	Number	Answer
1	62	2	B
3	D	4	C
5	A	6	B
7	C	8	A
9	A	10	C
11	13/2;6.5	12	B
13	A	14	C
15	B	16	A
17	B	18	A
19	D	20	15
21	1/72	22	C
23	A		

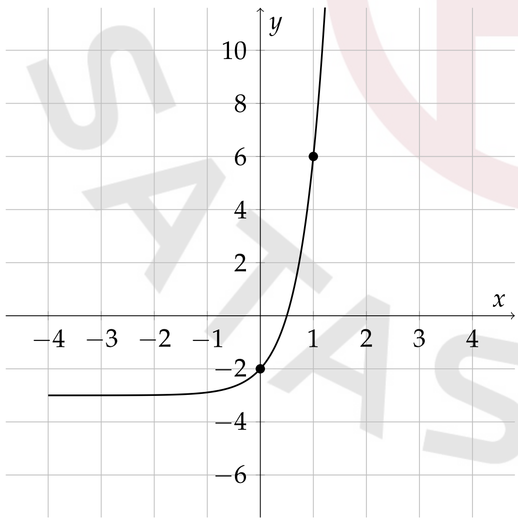
Exponential Functions

1. [September 2025]

Anton is selling toys from his toy collection. When he started selling the toys, there were 2,800 toys in the collection. At the end of each week for 3 weeks after Anton started selling the toys, the number of toys in the collection had decreased by 1/2 of the toys in the collection at the end of the previous week. Which equation gives the number of toys, c, in the collection w weeks after Anton started selling the toys, where $0 \leq w \leq 36$.

A) $c = w^{\frac{1}{2}}$
B) $c = 2800^w$
C) $c = 1400(\frac{1}{2})^w$
D) $c = 2800(\frac{1}{2})^w$

2. [August 2025]



The graph of the function f is shown, where $y = f(x)$. What is the value of $f(0)$?

A) 6
B) 2
C) -2
D) -6

3. [August 2025]

$$P(t) = 1,200(1.06)^t$$

The function P gives the estimated number of sea lions in a certain area, where t is the number of years since a study began. What is the best interpretation of $P(0) = 1,200$ in this context?

A) The estimated number of sea lions in the area increased by 1,200 each year during the study.
B) The estimated number of sea lions in the area increased by 106 each year during the study.
C) The estimated number of sea lions in the area was 1,200 when the study began.
D) The estimated number of sea lions in the area was 106 when the study began.

4. [September 2025]

$$f(x) = 20(2)^x + 6(2)^x$$

The function f is defined by the given equation. Which of the following equivalent forms of f displays, as a coefficient or a base, the y-coordinate of the y-intercept of the graph of $y = f(x)$ in the xy-plane?

A) $f(x) = 80(2)^{x-2} + 24(2)^{x-2}$
B) $f(x) = 2(10(2)^x + 3(2)^x)$
C) $f(x) = 20(4)^{\frac{x}{2}} + 6(4)^{\frac{x}{2}}$
D) $f(x) = 26(2)^x$

5. [August 2025]

The function f is defined by $f(x) = a^x - b$, where a and b are constants. In the xy-plane, the graph of $y = f(x)$ passes through the points $(c, 4)$ and $(2c, 114)$, where c is a constant. Which of the following could be the value of b?

A) 7

B) 14

C) 110

D) 118

6. [August 2025]

The function f is defined by $f(x) = a^x - b$, where a and b are constants. In the xy-plane, the graph of $y = f(x)$ passes through the points $(c, 5)$ and $(2c, 137)$, where c is a constant. Which of the following could be the value of b?

A) 7

B) 16

C) 132

D) 142

7. [August 2025]

$$P(t) = 1,900(1.07)^t$$

The function P gives the estimated number of sea lions in a certain area, where t is the number of years since a study began. What is the best interpretation of P(0) = 1,900 in this context?

A) The estimated number of sea lions in the area increased by 1,900 each year during the study.

B) The estimated number of sea lions in the area increased by 107 each year during the study.

C) The estimated number of sea lions in the area was 1,900 when the study began.

D) The estimated number of sea lions in the area was 107 when the study began.

8. [August 2025]

The function f is defined by $f(x) = 5(4)^{\frac{-x}{6}}$. What is the value of $f(12)$?

A) -80

B) -40

C) $-\frac{1}{400}$

D) $\frac{5}{16}$

9. [August 2025]

The number of bacteria in a growth medium is expected to increase by 160% every 2 hours during a period of observation. The number of bacteria in the growth medium was estimated to be 8,000 when the period of observation began. Which function P gives the expected number of bacteria in this growth medium t hours after the period of observation began?

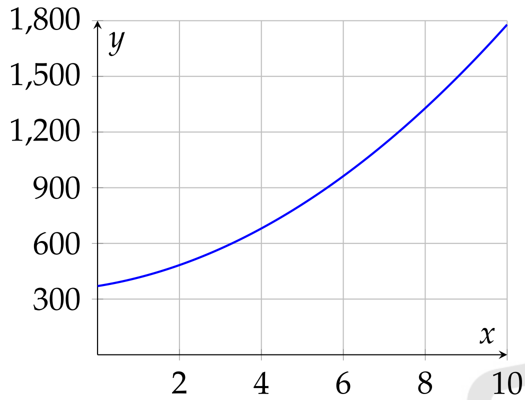
A) $P(t) = 8000(1.60)^{\frac{t}{2}}$

B) $P(t) = 8000(1.60)^{2t}$

C) $P(t) = 8000(2.60)^{\frac{t}{2}}$

D) $P(t) = 8000(2.60)^{2t}$

10. [August 2025]



The graph shows the estimated number of cell phone subscribers, y, in a town x years since 1990. Which equation represents the exponential relationship between the number of cell phone subscribers in the town and the number of years since 1990, where $0 \leq x \leq 10$?

A) $y = 370(0.17)^x$

B) $y = 370(1.17)^x$

C) $y = 1779(0.83)^x$

D) $y = 1779(1.83)^x$

11. [September 2025]

A researcher observes a sample of a nuclide. An exponential model estimates that the mass, in grams, of the sample decreases by 23% every 21.87 minutes. Which of the following equations could represent this model, where M is the estimated mass, in grams, of the sample t minutes after the researcher began observing the sample?

A) $M = 100(0.77)^{\frac{t}{21.87}}$

B) $M = 100(0.23)^{\frac{t}{21.87}}$

C) $M = 100(0.23)^{t+21.87}$

D) $M = 100(0.77)^{t+21.87}$

12. [September 2025]

$$g(x) = \frac{1}{11}(5(2)^x + 4)$$

The function g is defined by the given equation. For all values of x, the value of $g(x)$ is greater than k, where k is a constant. What is the greatest possible value of k?

A) 4

B) $\frac{4}{1}$

C) 9

D) $\frac{9}{11}$

13. [September 2025]

A beaker containing a liquid is placed on a table. The function $g(t) = 295 + (364 - 295)(2.72)^{-0.104t}$ gives the approximate temperature, in kelvins, of the liquid t minutes after the beaker was placed on the table. According to this function, what was the approximate temperature, in kelvins, of the liquid when the beaker was placed on the table?

14. [September 2025]

The function f is defined by $f(x) = a^x + b$, where a and b are constants and $a > 0$. In the xy-plane, the graph of $y = f(x)$ has a y-intercept at (0,-22) and passes through the point (2,41). What is the value of $a + b$?

15. [September 2025]

The population of a certain city doubled every 75 years from 1651 to 1951. The population of this city was 160,000 in 1951. What was the population of this city in 1651?

16. [September 2025]

A certain investment account offers a special interest rate for the first 4 months the account is open followed by a lower interest rate for the remainder of the time the account is open. Bennett opened one of these accounts with an original account balance of \$900 and did not make any other deposits or withdrawals. 4 months after Bennett opened the account, the balance had increased by 0.6% of the original balance. 6 months after Bennett opened the account, the balance had increased by an additional 0.1% of the balance at the end of the first 4 months. Every 2 months after the first 6 months, the balance had increased by an additional 0.1% of the balance 2 months before. Which of the following equations could represent the account balance $B(x)$, in dollars, x months after the account was opened, where $x \geq 4$?

A) $B(x) = 905.40(1.001)^{2x-8}$

B) $B(x) = 905.40(1.001)^{\frac{x}{2}-\frac{4}{2}}$

C) $B(x) = 905.40(1.001)^{\frac{x}{2}-4}$

D) $B(x) = 905.40(1.001)^{2x-4}$

17. [September 2025]

A beaker containing a liquid is placed on a table. The function $g(t) = 297 + (363 - 297)(2.72)^{-0.103t}$ gives the approximate temperature, in kelvins, of the liquid t minutes after the beaker was placed on the table. According to this function, what was the approximate temperature, in kelvins, of the liquid when the beaker was placed on the table?

18. [September 2025]

The function f is defined by $f(x) = a^x + b$, where a and b are constants and $a > 0$. In the xy-plane, the graph of $y = f(x)$ has a y-intercept at (0, -22) and passes through the point (2, 26). What is the value of $a + b$?

19. [September 2025]

The population of a certain city doubled every 75 years from 1657 to 1957. The population of this city was 240,000 in 1957. What was the population of this city in 1657?

20. [September 2025]

A certain investment account offers a special interest rate for the first months the account is open followed by a lower interest rate for the remainder of the time the account is open. Bennett opened one of these accounts with an original account balance of \$600and did not make any other deposits or withdrawals. 6 months after Bennett opened the account, the balance had increased by 0.4% of the original balance. 8 months after Bennett opened the account, the balance had increased by an additional 0.3% of the balance at the end of the first 6 months. Every 2 months after the first 8 months, the balance had increased by an additional of 0.3% the balance 2 months before. Which of the following equations could represent the account balance $B(x)$, in dollars, x months after the account was opened, where $x \geq 6$?

A) $B(x) = 602.40(1.003)^{\frac{x}{2}-6}$

B) $B(x) = 602.40(1.003)^{\frac{x}{2}-3}$

C) $B(x) = 602.40(1.003)^{2x-12}$

D) $B(x) = 602.40(1.003)^{2x-6}$

21. [September 2025]

What is the y-intercept of the graph of $y = 6^x + 12$ in the xy-plane?

A) $(0, 6)$

B) $(0, 12)$

C) $(0, 18)$

D) $(0, 13)$

22. [September 2025]

A researcher observes a sample of a nuclide. An exponential model estimates that the mass, in grams, of the sample decreases by 27% every 14.07 minutes. Which of the following equations could represent this model, where M is the estimated mass, in grams, of the sample t minutes after the researcher began observing the sample?

A) $M = 100(0.27)^{\frac{t}{14.07}}$

B) $M = 100(0.27)^{t+14.07}$

C) $M = 100(0.73)^{\frac{t}{14.07}}$

D) $M = 100(0.73)^{t+14.07}$

23. [September 2025]

$$g(x) = \frac{1}{11}(6(2)^x + 7)$$

The function g is defined by the given equation. For all values of x, the value of $g(x)$ is greater than k, where k is a constant. What is the greatest possible value of k?

A) 0

B) $\frac{13}{11}$

C) $\frac{7}{11}$

D) 7

24. [October 2025]

The y-intercept of the graph of $y = 6(1.85)^x$ in the xy-plane is $(0, y)$. What is the value of y?

25. [October 2025]

A model estimates that the initial number of photons in an X-ray beam is 800 when the beam reaches the surface of a certain material. The model also estimates that when the beam passes through this material, the number of photons in the beam decreases by 50% for each 7 millimeters of the material the beam passes through. Which equation represents this model, where I is the estimated number of photons in the beam when the beam reaches a point a millimeters from the surface?

A) $I = 800(7)^{\frac{x}{2}}$

B) $I = 800(0.5)^x$

C) $I = 800(2)^{\frac{x}{7}}$

D) $I = 800(0.5)^{\frac{x}{7}}$

26. [October 2025]

The function f is defined by $f(x) = 2x^3 + \frac{1}{3}x^2 - x$. What is the value of $f(1)$?

A) $\frac{4}{3}$

B) 2

C) $\frac{10}{3}$

D) 6

27. [October 2025]

The function f is defined by $f(x) = a(4)^x$, where a is a positive constant. In the xy-plane, the graph of $y = f(x)$ passes through the point (0,13). What is the value of $f(0)$?

A) 4

B) 1

C) 13

D) 9

28. [October 2025]

$$f(x) = 4000(0.90)^x$$

A conservation scientist implemented a program to reduce a certain invasive insect population in an area. The given function estimates this insect species' population x years after 2008, where $x \leq 8$. Which of the following is the best interpretation of 4000 in this context?

A) The estimated percent decrease in the insect population for this species and area every 8 years after 2008

B) The estimated initial insect population for this species and area in 2008

C) The estimated percent decrease in the insect population for this species and area each year after 2008

D) The estimated insect population for this species and area 8 years after 2008

29. [October 2025]

For $x = 0, y$ has a value of 200. The value of y increases by 40% for every increase of 1 in the value of x. Which equation represents this situation?

A) $y = 200(x)^{40}$

B) $y = 200(1.40)^x$

C) $y = 200(40)^x$

D) $y = 200(x)^{1.40}$

30. [October 2025]

The function f is defined by $f(x) = b^x + c$, where b and c are positive constants. In the xy-plane, the graph of $y = f(x)$ passes through the point $(0, 25)$. What is the value of $f(0)$?

31. [October 2025]

The relationship between the variables x and y is defined by an exponential equation. When $x = 0$, the value of y is 40, and for every increase in the value of x by 1, the corresponding value of y increases by 50% of its previous value. Which equation represents this relationship?

A) $y = 40(1.50)^x$

B) $y = 40(1.05)^x$

C) $y = 50(1.40)^x$

D) $y = 50(1.04)^x$

32. [October 2025]

$$y = 9(\frac{a}{7})^{x+c} - b$$

How many times does the graph of the given equation in the xy-plane cross the x-axis, where a, b, and c are positive constants such that $a > 7$ and $b < c$?

A) Zero

B) One

C) Two

D) Three

33. [October 2025]

The function h is defined by $h(x) = a^x + b$, where a and b are positive constants. The graph of $y = h(x)$ in the xy-plane passes through the points $(0, 10)$ and $(2, 13)$. What is the value of ab?

A) 13

B) 18

C) 20

D) 26

34. [October 2025]

The function f is defined by $f(x) = 56(0.19)^x$. For any positive integer n, the value of $f(n)$ is $p\%$ less than the value of $f(n-1)$. What is the value of p?

A) 19

B) 44

C) 56

D) 81

Answers: Exponential Functions

Number	Answer	Number	Answer
1	D	2	C
3	C	4	D
5	A	6	A
7	C	8	D
9	C	10	B
11	A	12	B
13	364	14	-15
15	10000	16	B
17	363	18	-16
19	15000	20	B
21	D	22	C
23	C	24	6
25	D	26	A
27	C	28	B
29	B	30	25
31	A	32	B
33	B	34	D

Quadratics

1. [September 2025]

$$\frac{1}{cx} = \frac{x}{96} + \frac{1}{c}$$

In the given equation, c is a constant. If the equation has exactly one solution, what is the value of c?

2. [August 2025]

$$x^2 - 9x + 1 = 0$$

What is the one solution to the given equation?

A) $\frac{9+\sqrt{77}}{2}$

B) $\frac{9+\sqrt{85}}{2}$

C) $\frac{-9+\sqrt{77}}{2}$

D) $\frac{-9+\sqrt{85}}{2}$

3. [August 2025]

For a certain type of rope, the equation $y = 900ax^2$, where a is a constant, gives the estimated breaking strength y, in pounds, of a rope with a circumference of x inches. Based on this equation, if a rope of this type has a circumference of 2.75 inches, it has an estimated breaking strength of 9,528.75 pounds. What is the estimated breaking strength, in pounds, of a rope of this type that has a circumference of 8.50 inches?

4. [August 2025]

$$(x + k)^2 + 31 = 31$$

In the given equation, k is a constant. How many distinct real solutions does this equation have?

A) Zero

B) Exactly one

C) Exactly two

D) Infinitely many

5. [August 2025]

$$f(x) = 4x^2 + 56x + 197$$

The function g is defined by $g(x) = f(x + 6)$. What is the minimum value of g(x)?

A) -13

B) -7

C) 1

D) 7

6. [August 2025]

$$(x - k)^2 = (k - 4a)(x - k)$$

In the given equation, a and k are constants, where $k > 4a$ The sum of the solutions to the equation is $3k + 37$ What is the value of a?

7. [September 2025]

If $(\frac{1}{2}x - \frac{21}{2})^2 - 81 = 0$, what is the value of $(x-21)^2$?

A) 324
B) 162
C) 21
D) 9

8. [September 2025]

$$y = 7x^2 - bx - 6$$

Which of the following equations is equal to the given equation above, where b is a positive constant.

A) $y = 7(x - \frac{b}{14})^2 - 6 - \frac{b^2}{28}$
B) $y = 7(x - \frac{b}{14})^2 - 6$
C) $y = 7(x + \frac{b}{14})^2 - 6 - \frac{b^2}{28}$
D) $y = 7(x + \frac{b}{14})^2 - 6$

9. [September 2025]

$$-x^2 + bx - 49 = 0$$

In the given equation, b is an integer. The equation has no real solution. What is least possible value of b?

10. [August 2025]

$$y - x = 46$$

$$y = x^2 - 44x$$

The graphs of the equations in the given system intersect at the point (x, y) in the xy-plane. What is a possible value of y?

A) 46
B) 47
C) 91
D) 92

11. [September 2025]

$$-5x^2 - x + 9 = 0$$

What is the greatest solution to the given equation?

A) $-\frac{1}{5} + \frac{\sqrt{181}}{5}$
B) $-\frac{1}{10} + \frac{\sqrt{181}}{10}$
C) $-\frac{1}{5} - \frac{\sqrt{181}}{5}$
D) $-\frac{1}{10} - \frac{\sqrt{181}}{10}$

12. [September 2025]

If $(x+2)^2 = 28$, what is the value of $x^2 + 4x$?

A) 56
B) 28
C) 32
D) 24

13. [September 2025]

$$\frac{1}{78}x^2 + (s - \frac{1}{78}t)x - st = 0$$

In the given equation, s and t are positive constants. The product of the solutions to the given equation is $-2kst$, where k is a constant. What is the value of k?

14. [September 2025]

A quadratic function gives the estimated length of daylight $d(t)$, in hours, in a certain city t months after March 1, where $0 \leq t \leq 7$. According to the function, the estimated length of daylight is 13.44 hours 5 months after March 1 and the maximum estimated length of daylight is 13.89 hours 3.5 months after March 1. Based on this function, what is the estimated length of daylight, in hours, on March 1?

15. [September 2025]

$$kx^2 - 16x = 27x^2 - 8$$

In the given equation, k is an integer constant. If the equation has two distinct real solutions, what is the greatest possible value of k?

16. [September 2025]

The expression $x^2 + kx + 14$, where k is a constant, can be rewritten as $(x+n)(x+7)$, where n is a constant. What is the value of k?

A) 2

B) 5

C) 9

D) 7

17. [September 2025]

$$-5x^2 - x + 9 = 0$$

What is the greatest solution to the given equation?

A) $-\frac{1}{5} + \frac{\sqrt{181}}{5}$

B) $-\frac{1}{10} + \frac{\sqrt{181}}{10}$

C) $-\frac{1}{5} - \frac{\sqrt{181}}{5}$

D) $-\frac{1}{10} - \frac{\sqrt{181}}{10}$

18. [September 2025]

In the xy-plane, the graph of the equation $y = -x^2 + 7x - 104$ intersects the line $y = c$ at exactly one point. What is the value of c?

A) $-\frac{367}{4}$

B) $-\frac{7}{2}$

C) -104

D) $-\frac{465}{4}$

19. [September 2025]

The expression $x^2 + kx + 55$, where k is a constant, can be rewritten as $(x + n)(x + 11)$, where n is a constant. What is the value of k?

A) 16

B) 6

C) 11

D) 5

20. [September 2025]

$$-3x^2 - 5x + 6 = 0$$

What is the greatest solution to the given equation?

A) $\frac{-5}{6} - \frac{\sqrt{97}}{6}$

B) $\frac{-5}{3} + \frac{\sqrt{97}}{3}$

C) $\frac{-5}{6} + \frac{\sqrt{97}}{6}$

D) $\frac{-5}{3} - \frac{\sqrt{97}}{3}$

21. [September 2025]

$$x^2 + 49x + 1 = 0$$

What is one of the solutions to the given equation?

A) $\dfrac{-49 + \sqrt{(49)^2 - 4(1)(1)}}{2(1)}$

B) $\dfrac{-49 + \sqrt{(49)^2 - 4(49)(49)}}{2(49)}$

C) $\dfrac{-1 + \sqrt{(1)^2 - 4(1)(49)}}{2(1)}$

D) $\dfrac{-1 + \sqrt{(1)^2 - 4(49)(1)}}{2(49)}$

22. [September 2025]

$$\frac{1}{24}x^2 + (s - \frac{1}{24}t)x - st = 0$$

In the given equation, s and t are positive constants. The product of the solutions to the given equation is $-2kst$, where k is a constant. What is the value of k?

23. [September 2025]

$$f(x) = -16x^2 + 76$$

The function f gives the estimated height, in feet, of an acorn x seconds after the acorn fell from a tree. Based on the function, what is the estimated height, in feet, of the acorn before it fell from the tree?

24. [September 2025]

$$\frac{(x-7)(x-8)}{x-3} = 0$$

What is the sum of the solutions to the given equation?

A) 8

B) 18

C) 15

D) 3

25. [September 2025]

$$-6x + 42px = 84$$

In the given equation, p is a constant. The equation has no solution. What is the value of p?

A) $\frac{1}{3}$

B) 0

C) 2

D) $\frac{1}{7}$

26. [September 2025]

If $(x+2)^2 = 18$, what is the value of $x^2 + 4x$?

A) 14

B) 22

C) 36

D) 18

27. [September 2025]

A quadratic function gives the estimated length of daylight $d(t)$, in hours, in a certain city t months after March 1, where $0 \leq t \leq 7$. According to the function, the estimated length of daylight is 12.66 hours 6 months after March 1 and the maximum estimated length of daylight is 13.91 hours 3.5 months after March 1. Based on this function, what is the estimated length of daylight, in hours, on March 1?

28. [September 2025]

$$nx^2 - 16x = 26x^2 - 8$$

In the given equation, n is an integer constant. If the equation has two distinct real solutions, what is the greatest possible value of n?

29. [October 2025]

A quadratic function models the height, in feet, of an object above the ground in terms of the time, in seconds, after the object was launched. The model estimates that the object was launched from an initial height of 0.00 feet above the ground and was at a maximum height of 207.36 feet above the ground 3.6 seconds after it was launched. The model also estimates that the object was at a height of 184.32 feet above the ground x seconds after it was launched. What is a possible value of x?

30. [October 2025]

$$x^2 + 49x + 1 = 0$$

A) $\dfrac{-49 + \sqrt{(49)^2 - 4(1)(1)}}{2(1)}$

B) $\dfrac{-49 + \sqrt{(49)^2 - 4(49)(49)}}{2(49)}$

C) $\dfrac{-1 + \sqrt{(1)^2 - 4(1)(49)}}{2(1)}$

D) $\dfrac{-1 + \sqrt{(1)^2 - 4(49)(1)}}{2(49)}$

31. [October 2025]

$15x^2 = 15(9)$

What is the negative solution to the given equation?

32. [October 2025]

The y-intercept of the graph of $y = 5x^2 + 72$ is $(0, y)$. What is the value of y?

33. [October 2025]

$$80x^2 + 80x = -10$$

One solution to the given equation can be written as $x = \dfrac{-8 - k\sqrt{2}}{16}$, where k is a constant. What is the value of k?

A) 4
B) 80
C) 32
D) 128

34. [October 2025]

$$k^2 - 47 = 97$$

What is the positive solution to the given equation?

A) 144
B) 12
C) 72
D) 50

35. [October 2025]

What is the y-intercept of the graph of $y = x^2 + 27$ in the xy-plane?

A) $(0, -27)$
B) $(0, 28)$
C) $(0, 27)$
D) $(0, 0)$

36. [October 2025]

For a certain type of rope, the equation $y = 900ax^2$, where a is a constant, gives the estimated breaking strength y, in pounds, of a rope with a circumference of x inches. Based on this equation, if a rope of this type has a circumference of 1.75 inches, it has an estimated breaking strength of 3,858.75 pounds. What is the estimated breaking strength, in pounds, of a rope of this type that has a circumference of 3.50 inches?

37. [October 2025]

$$x^2 - \frac{81}{16} = 0$$

How many distinct real solutions does the given equation have?

A) Zero
B) Exactly one
C) Exactly two
D) Infinitely many

38. [October 2025]

$$h(t) = -16t^2 + b$$

The function h estimates an object's height, in feet, above the ground t seconds after the object is dropped, where b is a constant. The function estimates that the object is 19.36 feet above the ground when it is dropped at $t = 0$. How many seconds after being dropped does the function estimate the object will hit the ground?

39. [October 2025]

$$x^2 + (\sqrt{k-3})x + 42 = 0$$

In the given equation, k is a constant. The equation has exactly one real solution. What is the value of k?

A) 171
B) 168
C) 165
D) 45

40. [October 2025]

$$9x^2 + 8 = nx$$

In the given equation, n is a constant. The equation has exactly one solution. What is the value of $\frac{n^2}{8}$

Answers: Quadratics

Number	Answer	Number	Answer
1	-24	2	A
3	91035	4	B
5	C	6	-37/4
7	A	8	A
9	-13	10	D
11	B	12	D
13	39	14	11.44; 286/25
15	34	16	C
17	B	18	A
19	A	20	C
21	A	22	12
23	76	24	C
25	D	26	A
27	11.46	28	33
29	2.4; 4.8	30	A
31	-3	32	72
33	A	34	B
35	C	36	15435
37	C	38	1.1;11/10
39	A	40	36

Problem solving

Problem Solving Formulas

- **Mean (Average)** $= \dfrac{\text{Sum of values}}{\text{Total number of values}}$.
- **Mode -** The most frequently occurring value.
- **Range -** The difference between the highest and lowest values in a dataset. **max-min**
- **Median -** The middle value in a sorted dataset, which divides the data into two equal halves.
- **Low standard deviation:** Data points are close to the mean.
- **High standard deviation:** Data is spread over a wide range.
- **Margin of Error:** How It Changes

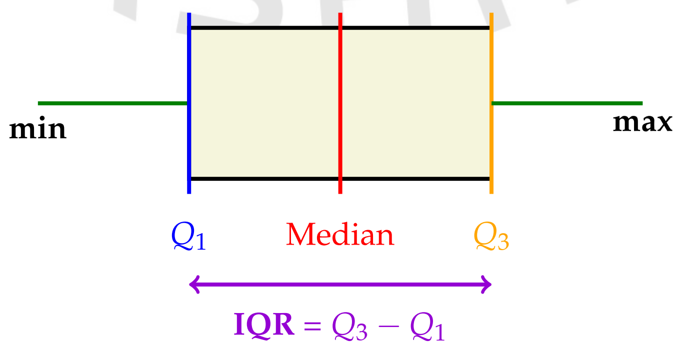
  Decreases When:

  - **Bigger Sample Size:** More data reduces uncertainty.
  - **Variability Decreases:** Less spread in the data narrows the error range.

  Increases When:

  - **Smaller Sample Size:** Less data increases uncertainty.
  - **Variability Increases:** More spread in the data enlarges the margin.

- **Probability** $= \dfrac{\text{desired outcomes}}{\text{total possible outcomes}}$.
- **Percent Change**$=\dfrac{\text{new}-\text{old}}{\text{old}} \cdot 100\%$.
- **Box Plot -** a way of summarizing data by showing:
  - The **minimum value** (left whisker).
  - The **lower quartile** (Q_1), which is the 25% point.
  - The **median** (50% point).
  - The **upper quartile** (Q_3), which is the 75% point.
  - The **maximum value** (right whisker).
  - The **Interquartile Range (IQR)** which is $Q_3 - Q_1$.



Percent; Ratio&Proportion

1. [September 2025]

An entomologist placed an initial population of 40 Tenebrio molitor, a type of beetle, into a habitat and monitored the population over time. When the number of Tenebrio molitor in the habitat reached 170% of the initial population, the entomologist moved 75% of the Tenebrio molitor to a second habitat. How many Tenebrio molitor did the entomologist move to the second habitat at this time?

2. [August 2025]

A certain bonefish can swim at a maximum speed of 17 meters per second. At this maximum speed, how many meters would this bonefish swim in 10 seconds?

3. [August 2025]

An object's speed is increasing at a rate of 6.80 meters per second squared. What is this rate in miles per minute squared, rounded to the nearest tenth? (Use 1 mile=1,609 meters.)

4. [September 2025]

The ratio 6 to 3 is equivalent to the ratio 24 to $3k$, where k is a constant. What is the value of k?

A) 0.75

B) 4

C) 12

D) 72

5. [August 2025]

On a plot of land, 48.0% of the square footage is farmland and the remaining square footage is pasture. There are buildings on exactly 23.5% of the square footage of the farmland, and there are buildings on exactly 16.0% of the square footage of the pasture. If there are buildings on exactly p% of the square footage of the plot of land, what is the value of p?

6. [September 2025]

For the positive quantities h, j, and k, 11% of h is equivalent to 33% of j, and j is equivalent to 14% of k. What percentage of k is h? (Disregard the % sign when entering your answer. For example, if your answer is 39%, enter 39)

7. [August 2025]

An object travels x inches in y seconds. At this rate, which expression represents the time, in seconds, it takes the object to travel a distance of $22x$ inches?

A) $\frac{22}{y}$

B) $\frac{y}{22}$

C) 22y

D) $22 + y$

8. [September 2025]

A certain goose species can fly at an average speed of 16 meters per second when in continuous flight. At this rate, how many meters would this goose species fly in 6 seconds?

A) 96

B) 16

C) 10

D) 22

9. [September 2025]

Stefan purchased 700 feet of fencing. He used 80% of this fencing to surround a vegetable garden. How many feet of fencing did Stefan use to surround the vegetable garden?

A) 40

B) 16

C) 560

D) 620

10. [September 2025]

A certain pigeon species can fly at an average speed of 16 meters per second when in continuous flight. At this rate, how many meters would this pigeon species fly in 3 seconds?

A) 16

B) 13

C) 48

D) 19

11. [August 2025]

For the positive quantities h, j, and k, 12% of h is equivalent to 95% of j, and j is equivalent to 24% of k. What percentage of k is h? (Disregard the % sign when entering your answer. For example, if your answer is 39%, enter 39)

12. [September 2025]

For the positive quantities h, j, and k, 13% of h is equivalent to 19% of j, and j is equivalent to of 91% of k . What percentage of k is h? (Disregard the % sign when entering your answer. For example, if your answer is 39% , enter 39)

13. [August 2025]

Stefan purchased 500 feet of fencing. He used 80% of this fencing to surround a vegetable garden. How many feet of fencing did Stefan use to surround the vegetable garden?

A) 12

B) 420

C) 400

D) 40

14. [October 2025]

There are a total of 340 seats in a school auditorium. During an assembly, students occupied 50% of the seats in the auditorium. How many seats did the students occupy during this assembly?

A) 290
B) 25
C) 170
D) 50

15. [October 2025]

The ratio of k to p is constant. If $k = 7$, then $p = 6$. What is the value of k when $p = 336$?

A) 288
B) 56
C) 48
D) 392

16. [October 2025]

Charles saves $\frac{3}{5}$ of the \$225 he earns each week from his summer job. If Charles continues to save at this rate, how much money, in dollars, will Charles save in 3 weeks?

17. [October 2025]

What is 10% of 450?

A) 35
B) 440
C) 405
D) 45

18. [August 2025]

The mass of object A is 444% of the mass of object B, and the mass of object A is 0.740% of the mass of object C. If the mass of object C is p% of the mass of object B, what is the value of p?

Answers: Percent, Ratio & Proportion

Number	Answer	Number	Answer
1	51	2	170
3	15.2	4	B
5	19.6	6	42
7	C	8	A
9	C	10	C
11	190	12	133
13	C	14	C
15	D	16	405
17	D	18	60000

Unit Conversion

1. [September 2025]

A certain town has an area of 13,753,344 square yards. What is the area, in square miles, of this town? (1 mile = 1,760 yards)

A) 7,814.4
B) 2.11
C) 3,708.55
D) 4.44

2. [September 2025]

A dog's mass is 24 kilograms. What is the dog's mass, in grams? (1 kilogram = 1,000 grams)

3. [September 2025]

The area of a rectangular region is increasing at a rate of 210 square feet per hour. Which of the following is closest to this rate in square meters per minute? (Use 1 meter = 3.28 feet.)

A) 0.33
B) 1.07
C) 11.48
D) 19.52

4. [September 2025]

A certain town has an area of 12,669,184 square yards. What is the area, in square miles, of this town?(1 mile = 1,760 yards)

A) 2.02
B) 7198.4
C) 4.09
D) 3559.38

5. [September 2025]

The area of a rectangular region is increasing at a rate of 190 square feet per hour. Which of the following is closest to this rate in square meters per minute? (Use 1 meter = 3.28 feet.)

A) 10.39
B) 0.97
C) 0.29
D) 17.66

6. [October 2025]

A triathlon is a multisport race consisting of three different legs. A triathlon participant completed the cycling leg with an average speed of 17,700 miles per hour. What was the average speed, in million yards per hour, of the participant during the cycling leg? (1 mile = 1,760 yards)

7. [October 2025]

A triathlon is a multisport race consisting of three different legs. A triathlon participant completed the cycling leg with an average speed of 19.3 miles per hour. What was the average speed, in yards per hour, of the participant during the cycling leg? (1 mile = 1760 yards)

A) 33968

B) 91.2

C) 57.9

D) 5280

8. [October 2025]

A length of 450 meters is equal to how many decimeters? (1 meter = 10 decimeters)

9. [October 2025]

A certain town has an area of 4.29 square miles. What is the area, in square yards, of this town? (1 mile = 1760 yards)

A) 410

B) 7,550

C) 722,051

D) 13,288,704

Answers: Unit Conversion

Number	Answer	Number	Answer
1	D	2	24000
3	A	4	C
5	C	6	31.152
7	A	8	4500
9	D		

Probability

1. [August 2025]

Survey Results	
Annuals	159
Perennials	295

A total of 454 gardeners selected at random were asked if they preferred perennials or annuals. The results of the survey are shown in the table above. According to these results, if 5,448 gardeners were choosing plants, how many more would prefer perennials than would prefer annuals?

A) 136

B) 1632

C) 1908

D) 3540

2. [September 2025]

The table summarizes the distribution of age and assigned group for 90 participants in a study.

	0–9 years	10–19 years	20+ years	Total
Group A	15	11	4	30
Group B	4	5	21	30
Group C	11	14	5	30
Total	30	30	30	90

One of these participants will be selected at random. What is the probability of selecting a participant from group A, given that the participant is at least 10 years of age?

A) $\frac{1}{6}$

B) $\frac{1}{4}$

C) $\frac{11}{30}$

D) $\frac{1}{2}$

3. [September 2025]

A department store sells pants in three lengths: short, regular, and long. The table summarizes the distribution of pant length, by color, for all the pants in the store.

	Black	**Brown**	**Blue**	**Total**
Short	19	20	24	63
Regular	17	21	22	60
Long	14	25	19	58
Total	50	66	65	181

A pair of pants is selected at random. What is the probability of selecting a pair of pants that is brown, given that the pair of pants is regular length? (Express your answer as a decimal or fraction, not as a percent.)

4. [September 2025]

A department store sells pants in three lengths: short, regular, and long. The table summarizes the distribution of pant length, by color, for all the pants in the store.

	Black	Brown	Blue	Total
Short	25	14	24	63
Regular	23	15	22	60
Long	20	19	19	58
Total	68	48	65	181

A pair of pants is selected at random. What is the probability of selecting a pair of pants that is brown, given that the pair of pants is regular length? (Express your answer as a decimal or fraction, not as a percent.)

5. [October 2025]

Each of 185 animals can be classified as one of three species, as shown in the frequency table.

Classification	Frequency
species X	112
species Y	3
species Z	70

If one of the animals is selected at random, what is the probability of selecting an animal of species Y?

A) $\frac{70}{185}$
B) $\frac{3}{185}$
C) $\frac{182}{185}$
D) $\frac{112}{185}$

6. [October 2025]

The table shows the distribution of people in a certain city by age group.

Age group	Proportion
Less than 18 years old	26%
18-40 years old	21%
41-65 years old	29%
Greater than 65 years old	24%

If a person in this city is selected at random, which of the following is closest to the probability of selecting a person who is greater than 65 years old, given that the person is at least 18 years old?

A) 0.24
B) 0.32
C) 0.50
D) 0.92

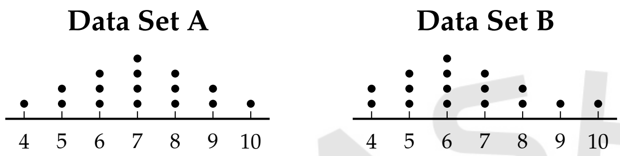
Answers: Probability

Number	Answer	Number	Answer
1	B	2	B
3	0.35;7/20	4	1/4;0.25
5	B	6	B

Mean, Median, Mode, Range

1. [September 2025]

The dot plots represent the distributions of values in data sets A and B.



Which of the following statements must be true?

I. The median of data set A is equal to the median of data set B.

II. The standard deviation of data set A is equal to the standard deviation of data set B.

A) I and II

B) I only

C) II only

D) Neither I nor II

2. [August 2025]

A farmer gave two groups of chickens different types of chicken feed for a month to measure the effect on egg production. The lists give the number of eggs collected from each chicken in each of the two groups.

Group G: 8, 16, 18, 18, 24
Group H: 16, 18, 18, 24

Which statement correctly compares the median number of eggs collected for group G and the median number of eggs collected for group H?

A) The median number of eggs collected for group G is equal to the median number of eggs collected for group H.

B) The median number of eggs collected for group G is greater than the median number of eggs collected for group H.

C) The median number of eggs collected for group G is less than the median number of eggs collected for group H.

D) There is not enough information to compare the median number of eggs collected for group G and the median number of eggs collected for group H.

3. [September 2025]

$$-3x + 15px = 30$$

In the given equatoin, p is a constant. The equation had no solution. What is the value of p?

A) $(\frac{2}{3}$

B) 0

C) 2

D) $\frac{1}{5}$

4. [September 2025]

At a nature preserve, a wildlife biologist counted ducks from an observation deck at the same time each day for 41 days. The table summarizes the resulting data set, data set A.

Number of ducks	Number of days
0	1
1	7
2	8
3	9
4	8
5	7
18	1

The data value 18 was recorded in error and is removed from data set A to create data set B, which consists of the remaining 40 data values. Which statement best compares the median of data set A and the median of data set B?

A) There is not enough information to compare the medians of the two data sets.

B) The median of data set B is equal to the median of data set A.

C) The median of data set B is greater than the median of data set A.

D) The median of data set B is less than the median of data set A.

5. [September 2025]

Each of the following frequency tables represents a data set. Which data set has the greatest mean?

A)

Value	Frequency
10	3
20	4
30	5
40	6

B)

Value	Frequency
10	7
20	4
30	4
40	7

C)

Value	Frequency
10	5
20	5
30	5
40	5

D)

Value	Frequency
10	6
20	5
30	5
40	6

6. [September 2025]

Each of the following frequency tables represents a data set. Which data set has the greatest mean?

A)

Value	*Frequency*
60	3
70	3
80	3
90	3

B)

Value	*Frequency*
60	4
70	3
80	3
90	4

C)

Value	*Frequency*
60	1
70	2
80	3
90	4

D)

Value	*Frequency*
60	5
70	2
80	2
90	5

7. [September 2025]

At a nature preserve, a wildlife biologist counted ducks from an observation deck at the same time each day for 41 days. The table summarizes the resulting data set, data set A.

Number of ducks	Number of days
0	1
1	7
2	8
3	9
4	8
5	7
13	1

The data value 13 was recorded in error and is removed from data set A to create data set B, which consists of the remaining 40 data values. Which statement best compares the median of data set A and the median of data set B?

A) The median of data set B is greater than the median of data set A.

B) There is not enough information to compare the medians of the two data sets.

C) The median of data set B is equal to the median of data set A.

D) The median of data set B is less than the median of data set A.

8. [October 2025]

For a study, a research team measured the heights of 380 king penguins from the Kerguelen Islands and 190 king penguins from the Falkland Islands. The research team determined that the mean height of the 380 king penguins from the Kerguelen Islands was 87 centimeters and the mean height of the 190 king penguins from the Falkland Islands was 93 centimeters. What was the mean height, in centimeters, of all 570 king penguins the research team measured for this study?

A) 91
B) 89
C) 90
D) 88

9. [October 2025]

The List give us the speeds, in miles per hour, of 7 servers made by a professional tennis player.

121, 123, 123, 125, 151, 154, 155

What is the mean speed, in miles per hour, of these 7 serves?

10. [October 2025]

$$a, 26, 29, b, 31, 47, c$$

For the given data set, the data values are listed in ascending order, where a, b, and c are constants. For this data set, the mean is 36, the median is 29, and the range is 72. What is the value of c?

A) 54
B) 72
C) 81
D) 98

11. [October 2025]

For data set A, the table summarizes the distribution of the number of pieces of mail received by a business each day during a period of 11 days.

Pieces of mail	Days
0	2
3	2
4	2
5	2
6	1
7	1
13	1

The data value 13 is removed from data set A to create data set B, which consists of the remaining 10 data values. Which statement best compares the median of data set A and the median of data set B?

A) The median of data set B is less than the median of data set A.
B) The median of data set B is greater than the median of data set A.
C) The median of data set B is equal to the median of data set A.
D) There is not enough information to compare the medians of the two data sets.

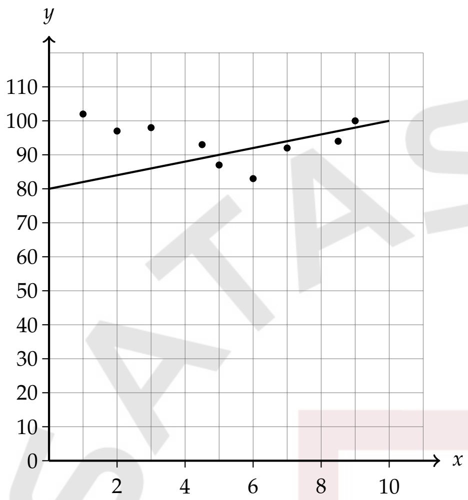
Answers: Mean, Median, Mode, Range

Number	Answer	Number	Answer
1	C	2	A
3	D	4	B
5	A	6	C
7	C	8	B
9	136	10	C
11	C		

Scatterplots

1. [September 2025]

The scatterplot shows the relationship between two variables, x and y. A line of best fit for the data is also shown.



For how many of the 9 data points is the actual y-value greater than the y-value predicted by the line of best fit?

A) 8

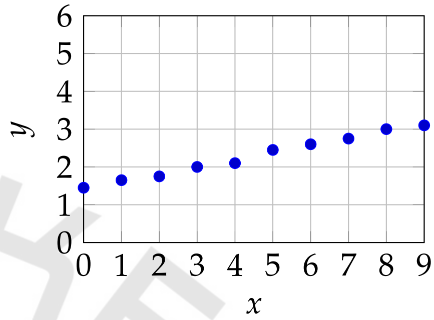
B) 7

C) 5

D) 2

2. [August 2025]

The graph shows the number of internet users worldwide y, in billion, x years after



2007.

Which of the following is closest to the number of internet users worldwide, in billions, 5 years after 2007?

A) 1.4

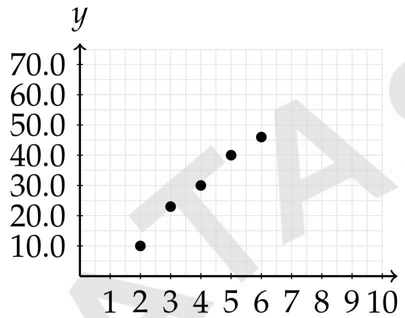
B) 2.4

C) 3.4

D) 4.4

3. [September 2025]

The scatterplot shows data set A, which consists of the weights y, in pounds, of a Labrador retriever puppy at various ages, x, in months. The equation of a line of best fit for the relationship in data set A can be written as $y = -5.1 + 8.7x$, where $2 \geq x \geq 6$

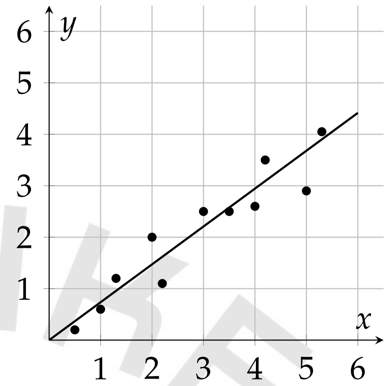


The puppy was weighed again at 9 months old and weighed 52 pounds. Data set B consists of all the data points in data set A as well as the data point (9, 52). The equation of a line of best fit for data set B can be written as $y = r + st$, where r and s are constants and $2 \geq x \geq 9$. Assuming the equations of the lines of best fit are calculated in the same way, which of the following is the best estimate for the value of s?

A) 5.9
B) 8.7
C) 13.8
D) 17.7

4. [September 2025]

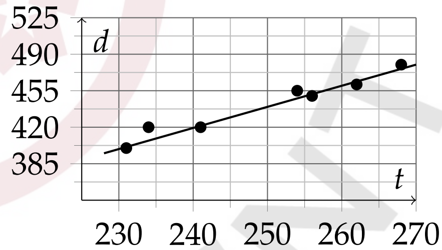
The scatterplot shows the relationship between two variables, x and y. A line of best fit is also shown.



Which of the following equations best represents the line of best fit shown?

A) $y = 0$
B) $y = 1.7$
C) 0.7
D) $y = 1.7 + 0.7x$

5. [October 2025]

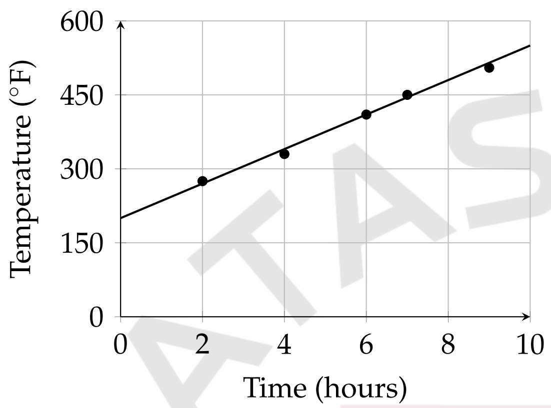


Which of the following equations is the most appropriate linear model for the data shown?

A) $d = -62.1 + 2.02t$
B) $d = 162.1 + 2.02t$
C) $d = 357.8 + 2.02t$
D) $d = 392.8 + 2.02t$

6. [October 2025]

For the first 10 hours of an experiment, the scatterplot shows the temperature y, in degrees Fahrenheit (°F), of an object at various times x, in hours, since the start of the experiment. A line of best fit is also shown.



Which of the following is the best interpretation of the slope of the line of best fit?

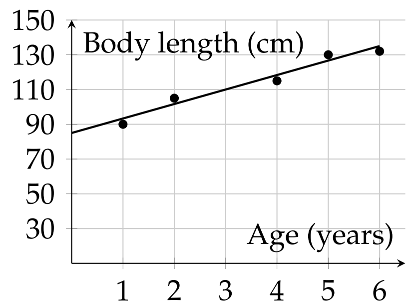
A) The predicted temperature increases at a constant rate of approximately 40°F per hour over the 10-hour period.

B) The predicted temperature decreases at a constant rate of approximately 40°F per hour over the 10-hour period.

C) The predicted temperature decreases at a constant rate of approximately 180°F per hour over the 10-hour period.

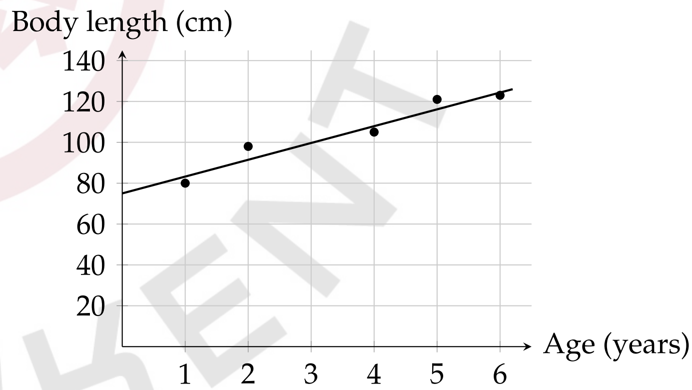
D) The predicted temperature increases at a constant rate of approximately 180°F per hour over the 10-hour period.

7. [September 2025]



The scatterplot shows 5 measurements of the body length, in centimeters (cm), of a New Zealand fur seal from an age of 1 year to 6 years old. A line of best fit is also shown. For a New Zealand fur seal at an age of 3 years old, what is the body length predicted by the line of best fit, to the nearest 10 cm?

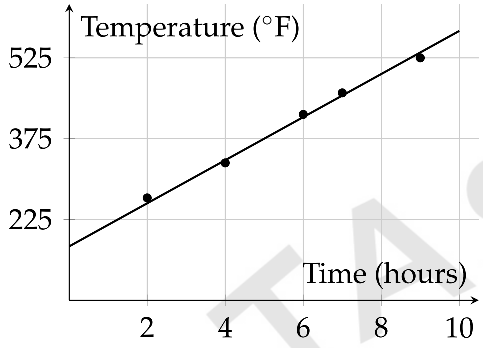
8. [September 2025]



The scatterplot shows 5 measurements of the body length, in centimeters (cm), of a New Zealand fur seal from an age of 1 year to 6 years old. A line of best fit is also shown. For a New Zealand fur seal at an age of 3 years old, what is the body length predicted by the line of best fit, to the nearest ?

9. [September 2025]

For the first 10 hours of an experiment, the scatterplot shows the temperature y, in degrees *Fahrenheit*(F), of an object at various times, x in hours, since the start of the experiment. A line of best fit is also shown.



Which of the following is the best interpretation of the slope of the line of best fit?

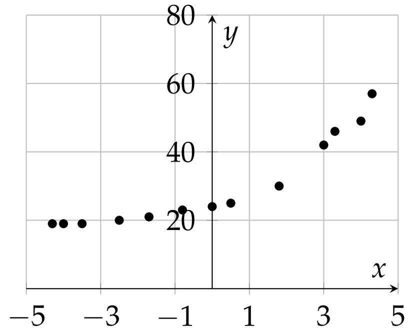
A) The predicted temperature increases at a constant rate of approximately 170°Fper hour over the 10-hour period.

B) The predicted temperature decreases at a constant rate of approximately 170°Fper hour over the 10-hour period.

C) The predicted temperature increases at a constant rate of approximately per hourover the 10-hour period.

D) The predicted temperature decreases at a constant rate of approximately 40°F perhour over the 10-hour period.

10. [October 2025]



- The scatterplot has 13 data points.
- The data points are in an exponential pattern trending up from left to right.
- The data points have the following approximate coordinates:

$(-4.3, 19)$ $(-4, 19)$ $(-3.5, 19)$

$(-2.5, 20)$ $(-1.7, 21)$ $(-0.8, 23)$

$(0, 24)$ $(0.5, 25)$ $(1.8, 30)$

$(3, 42)$ $(3.3, 46)$ $(4, 49)$

$(4.3, 57)$

Which of the following equations is the most appropriate model for the data shown in the scatterplot?

A) $y = 24(1.55)^x$

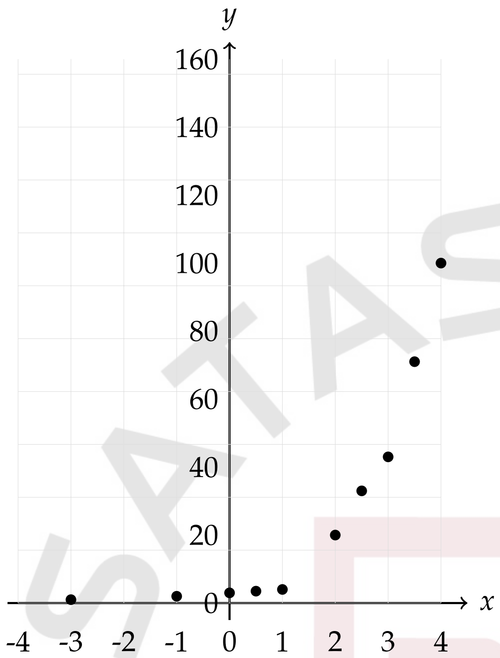
B) $y = 24(1.55)^x + 18$

C) $y = (1.55)^x + 23$

D) $y = 6(1.55)^x + 18$

11. [October 2025]

The scatterplot shows the relationship between x and y for the 10 data points in data set A.



Data set B is created by multiplying the y-value of each data point from data set A by 40. Which of the following equations is the most appropriate model for data set B?

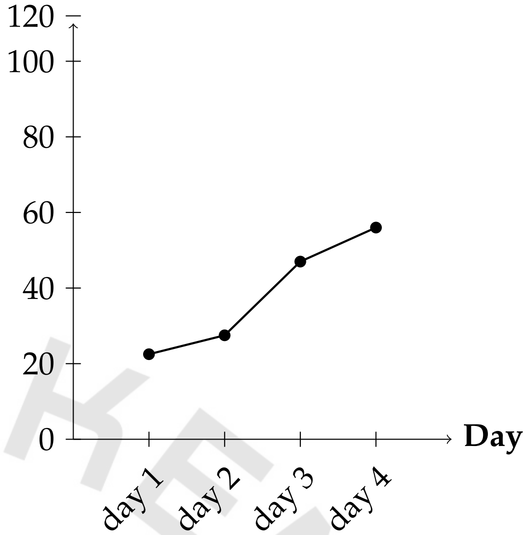
A) $y = 5(2.04)^x$

B) $y = 40(2.04)^x + 40$

C) $y = 5(2.04)^x + 40$

D) $y = 199(2.04)^x + 2.04$

12. [October 2025]



The line graph shows the number of customers who visited a restaurant each day during a four-day period in the summer. According to the line graph, for which day during this four-day period did the restaurant have the smallest number of customers who visited?

A) Day 1

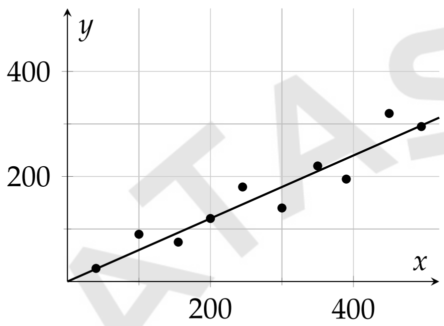
B) Day 2

C) Day 3

D) Day 4

13. [October 2025]

A group of 10 gardeners recorded data on the germination rates of their tomato crop for one growing season. The scatterplot shows the relationship between the number of tomato seeds planted, x, and the number of tomato seeds that germinated, y, for each of the gardeners. A line of best fit is also shown.



Which of the following is the best interpretation of the slope of the line of best fit in this context?

A) The number of tomato seeds planted is predicted to increase by 60 seeds every 100 days.

B) The number of tomato seeds planted is predicted to increase by 300 seeds every 100 days.

C) The number of tomato seeds that germinate is predicted to increase by 60 seeds for every additional 100 tomato seeds that are planted.

D) The number of tomato seeds that germinate is predicted to increase by 300 seeds for every additional 100 tomato seeds that are planted.

Answers: Scatterplots

Number	Answer	Number	Answer
1	C	2	B
3	A	4	D
5	A	6	A
7	110	8	100
9	C	10	D
11	D	12	A
13	C		

Research organizing(Margin of Error; Outliers)

1. [October 2025]

For a survey, a sample of 996 people was selected at random from all registered voters in a county. Based on the survey, it is estimated that 42% of all registered voters in the county are in favor of a new law, with an associated margin of error of 3.07%. Which of the following best describes the plausible values for the percent of all registered voters in the county who are in favor of the new law?

A) Any value greater than 38.93% and less than 45.07%

B) Any value greater than 38.93%

C) Any value less than 38.92% or greater than 45.07%

D) Any value less than 45.07%

2. [September 2025]

A science club from a certain high school in Ohio conducted an experiment to study the effects of smell on taste. A sample of 125 students was selected at random from the high school to participate in the experiment. First, the students tasted a pureed food sample while wearing covered goggles and nose plugs. Then, the students tasted the same pureed food sample without nose plugs. 50 students were able to correctly identify the pureed food sample while wearing nose plugs. Which of the following is the largest population to which the results of the experiment can be generalized?

A) The 50 students who were able to correctly identify the pureed food sample while wearing nose plugs

B) All students from the high school

C) The 125 students who participated in the experiment

D) All students from high schools in Ohio

3. [September 2025]

A science club from a certain high school in Ohio conducted an experiment to study the effects of smell on taste. A sample of 100 students was selected at random from the high school to participate in the experiment. First, the students tasted a pureed food sample while wearing covered goggles and nose plugs. Then, the students tasted the same pureed food sample without nose plugs. 35 students were able to correctly identify the pureed food sample while wearing nose plugs. Which of the following is the largest population to which the results of the experiment can be generalized?

A) All students from the high school

B) The 100 students who participated in the experiment

C) The 35 students who were able to correctly identify the pureed food sample while wearing nose plugs

D) All students from high schools in Ohio

4. [October 2025]

For certain air temperatures, the table gives wind chill temperatures for two different wind speeds.

Air temperature	Wind chill temperature at wind speed 5 mph	Wind chill temperature at wind speed 15 mph
27 °F	21 °F	15 °F
29 °F	24 °F	18 °F
31 °F	26 °F	20 °F

According to the table, what is the wind chill temperature, in degrees Fahrenheit (°F), when the air temperature is 26 Fahrenheit (°F) and the wind speed is 15 mph? (Disregard the degree when entering your answer)

5. [October 2025]

The frequency table summarizes the 105 data values in a data set.

Data value	Frequency
28	13
30	15
31	16
32	17
33	16
34	15
35	13

What is the minimum value of the data set?

6. [October 2025]

A snack company advertises that bags of pretzels produced by the company weigh, on average, 1 pound each. To test this, Sam selected at random 40 bags of pretzels produced by the company and weighed each bag. Based on his measurements, Sam estimated that for all bags of pretzels produced by the company, the average weight per bag is 1.05 pounds, with a margin of error of 0.06 pounds. For all bags of pretzels produced by the company, which of the following is the most appropriate conclusion about?

A) $0.99 \leq w \leq 1.05$

B) $w = 1.05$

C) $1.05 \leq w \leq 1.11$

D) $0.99 \leq w \leq 1.11$

Answers: Research Organizing

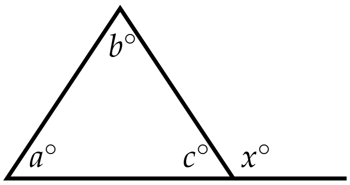
Number	Answer	Number	Answer
1	A	2	B
3	A	4	14
5	28	6	D

Geometry

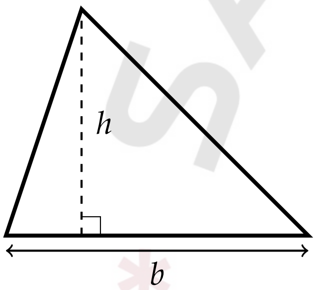
Geometry & Trigonometry Formulas

Triangles

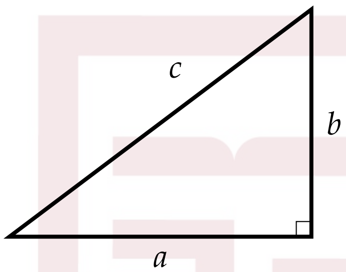
- The sum of the measures in degrees of the angles of a triangle is 180°.



An **exterior angle** is always equal to the sum of the two angles in the triangle furthest from it. In this case, $x = a + b$.

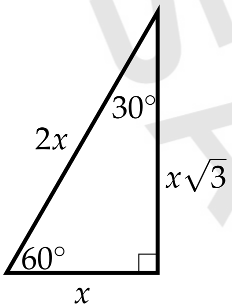


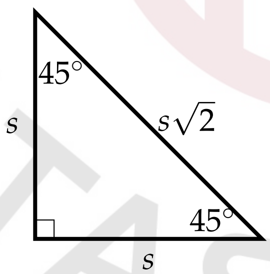
$A = \frac{1}{2}hb$



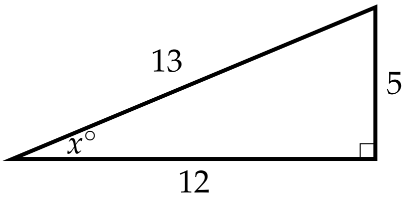
$c^2 = a^2 + b^2$

Special Right Triangles

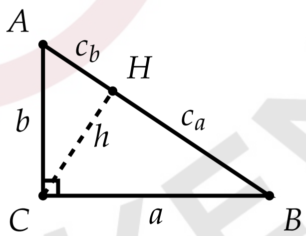




Trigonometry



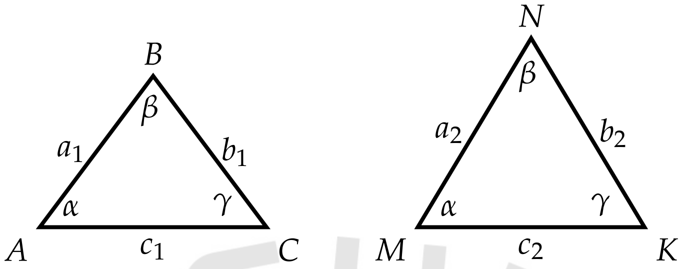
- $\sin x = \dfrac{\text{opposite}}{\text{hypotenuse}} = \dfrac{5}{13}$
- $\cos x = \dfrac{\text{adjacent}}{\text{hypotenuse}} = \dfrac{12}{13}$
- $\tan x = \dfrac{\text{opposite}}{\text{adjacent}} = \dfrac{5}{12}$
- $\mathbf{\sin x = \cos(\frac{\pi}{2} - x)}$
- $\mathbf{\cos x = \sin(\frac{\pi}{2} - x)}$
- $\cos x = \sin y$, when $x + y = 90^\circ$



- $h = \sqrt{c_a c_b}$
- $h = \dfrac{ab}{c}$
- CH−height=h
- $AH = c_b$
- $HB = c_a$

Similarity&Congruency

- Two figures (such as triangles) are **similar (∼)** if they have the same shape, regardless of their size. $\triangle ABC \sim \triangle MNK$

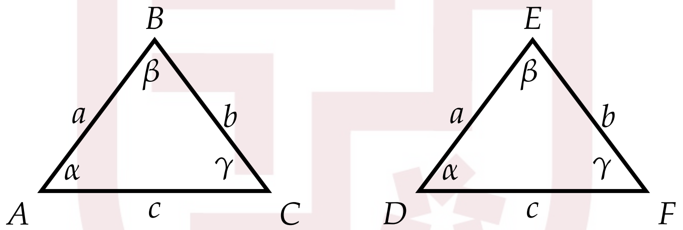


$$\frac{a_1}{a_2} = \frac{b_1}{b_2} = \frac{c_1}{c_2} = \frac{p_1}{p_2} = \frac{h_1}{h_2},$$

$$\left(\frac{a_1}{a_2}\right)^2 = \frac{A_1}{A_2},$$

If two 3D shapes are similar: $\left(\frac{a_1}{a_2}\right)^3 = \frac{V_1}{V_2}.$

- Two figures are **congruent (≅)** if they have exactly the same size and shape. This means that all corresponding sides and angles in the figures are equal.



- **SSS (Side-Side-Side):**
If all three sides of one triangle are equal to the corresponding three sides of another triangle, then the triangles are congruent.

$$\triangle ABC \cong \triangle DEF$$

When: $AB = DE,\ BC = EF,\ CA = FD.$

- **SAS (Side-Angle-Side):**
If two sides and the included angle of one triangle are equal to two sides and the included angle of another triangle, then the triangles are congruent.

$$\triangle ABC \cong \triangle DEF$$

When: $AB = DE, \quad \angle B = \angle E, \quad BC = EF.$

- **ASA (Angle-Side-Angle):**
If two angles and the included side of one triangle are equal to two angles and the included side of another triangle, then the triangles are congruent.

$$\triangle ABC \cong \triangle DEF$$

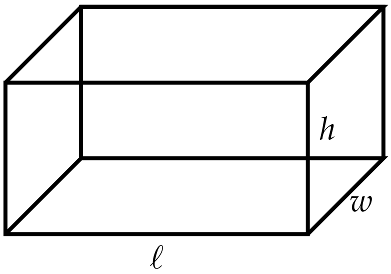
When: $\angle A = \angle D, \quad AB = DE, \quad \angle B = \angle E.$

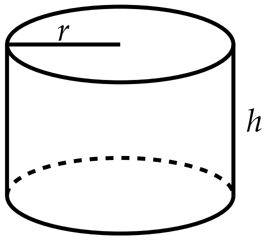
- **AAS (Angle-Angle-Side) or SAA (Side-Angle-Angle):**
  If two angles and a non-included side of one triangle are equal to the corresponding two angles and side of another triangle, then the triangles are congruent.

$$\triangle ABC \cong \triangle DEF$$

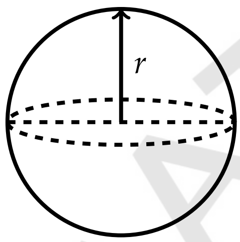
  When: $\angle A = \angle D, \quad \angle B = \angle E, \quad BC = EF.$

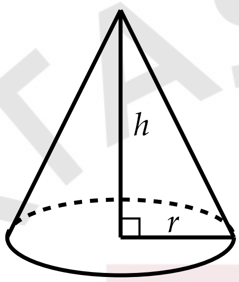
3D shapes



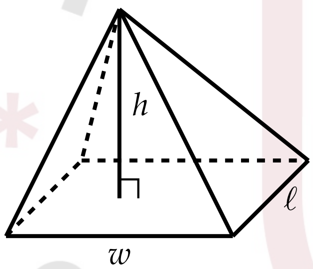


$V = \ell h w$ $\qquad$ $V = h\pi r^2$





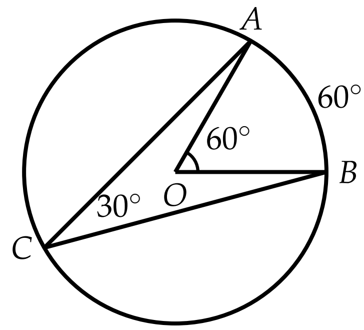
$V = \frac{4}{3}\pi r^3$ $\qquad$ $V = \frac{1}{3}\pi r^2 h$



$V = \frac{1}{3}wlh$

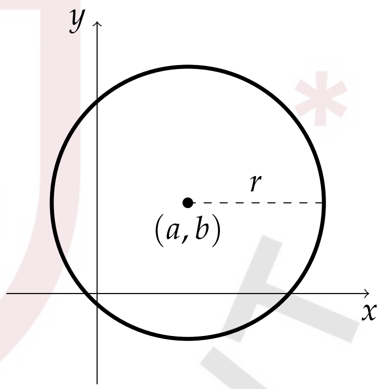
Circle

- The number of degrees of arc in a circle is 360°.
- The number of radians of arc in a circle is 2π.
- Sum of interior angles$=180^\circ(n-2)$



- OA—radius$=r$
- $\angle AOB$—central angle
- $\angle ACB$—inscribed angle
- **Arc length:** $\dfrac{\theta}{360} \cdot 2\pi r$
- **Area of sector:** $\dfrac{\theta}{360} \cdot \pi r^2$

θ— central angle in degrees.

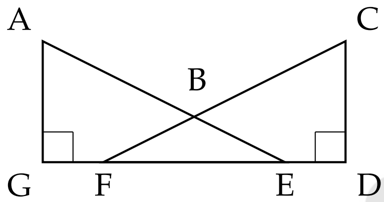


$$(x-a)^2 + (y-b)^2 = r^2$$

- r—radius
- (a, b)—center coordinate of circle
- Sum of exterior angles$=360^\circ$

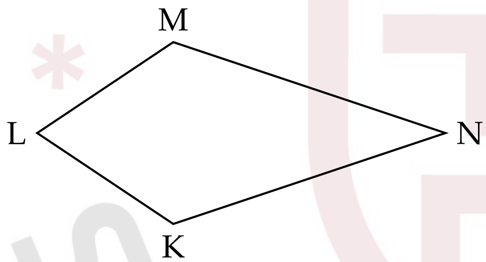
Lines and Angles

1. [August 2025]



In the figure, triangle AEG is congruent to triangle CFD, where E corresponds to F. The measure of angle CBE is $56°$. What is the measure in degrees , of angle EAG?

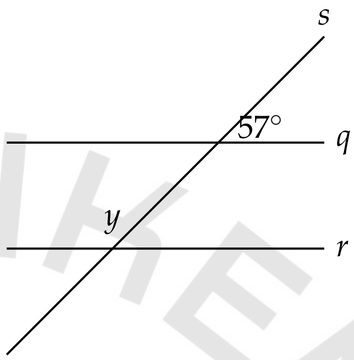
2. [August 2025]



In quadrilateral $KLMN$ shown, $KL = 3$, $LM = 3$, $KN = 27$, and $MN = 27$. Diagonals KM and LN (not shown) intersect at point G (not shown), where $GK = 1$ and $GM = 1$ if the length of diagonal LN is $\sqrt{p} + \sqrt{w}$, where p and w are integers, what is the value of $p + w$?

3. [August 2025]

In the figure, line q is parallel to line r, and both are intersected by line s. If $y = 2x + 7$, what is the value of x?



A) 25

B) 32

C) 58

D) 68

4. [August 2025]

The measure of angle G is $\frac{\pi}{10}$ radians. If the measure of angle G is $9n$ in degrees, where n is a constant, what is the value of n?

A) 2

B) 9

C) 10

D) 18

5. [August 2025]

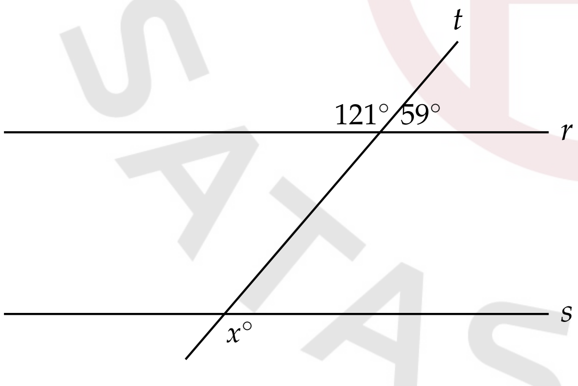
A line intersects two parallel lines, forming four acute angles and four obtuse angles. The measure of one of the acute angles is $(7x - 410)^\circ$. The sum of the measures of one of the acute angles and three of the obtuse angles is $(-14x + w)^\circ$. What is the value of w?

6. [August 2025]

The measure of angle S is $\frac{7\pi}{17}$ radians. The measure of angle T is 2 times the measure of angle S. Which expression represents the measure, in degrees, of angle T?

A) $\frac{7}{17}(900)(2)$
B) $\frac{7}{17}(180)(2)$
C) $\frac{7}{17\pi}(90)(2)$
D) $\frac{7\pi}{17}1802$

7. [August 2025]



Note: Figure not drawn to scale.

In the figure shown, line r is parallel to line s, and line t intersects both lines. What is the value of x?

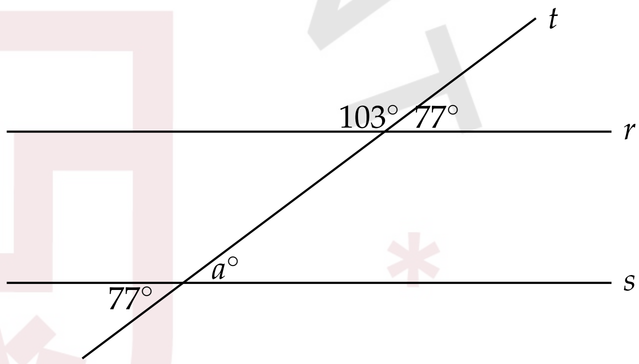
A) 121
B) 90
C) 31
D) 180

8. [August 2025]

A rectangle has a perimeter of 36 inches. If the sum of the lengths of three sides of the rectangle is 21 inches, what is the length, in inches, of the fourth side?

A) 3
B) 57
C) 15
D) 36

9. [September 2025]



Note: Figure not drawn to scale.

In the figure shown, lines r and s are parallel, and line t intersects both lines. What is the value of a?

A) 180
B) 77
C) 167
D) 90

10. [September 2025]

The measure of angle K is $\frac{\pi}{2(12)}$ radians. The measure of angle L is 12 times the measure of angle K. What is the measure, in degrees, of angle L?

A) 102

B) 12

C) 78

D) 90

11. [September 2025]

In triangle ABC, the measure of angle A is 52 and AC=40. In triangle PQR, the measure of angle P is 52 and PR=120 . Which additional piece of information is sufficient to prove that triangle ABC is similar to triangle PQR?

A) AB = 45 and PQ = 45.

B) AB = 45 and QR = 135.

C) The measures of angle B and angle R are 40° and 88°, respectively.

D) The measures of angle B and angle Q are 52° and 40°, respectively

12. [October 2025]

Triangle ABC is similar to triangle DEF, where angle A corresponds to angle D and angle C corresponds to angle F. Angles C and F are right angles. If $\tan A = \dfrac{612}{35}$, what is the value of $\sin D$?

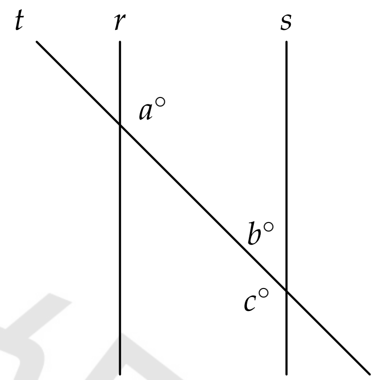
A) $\frac{612}{613}$

B) $\frac{35}{613}$

C) $\frac{613}{612}$

D) $\frac{613}{35}$

13. [October 2025]



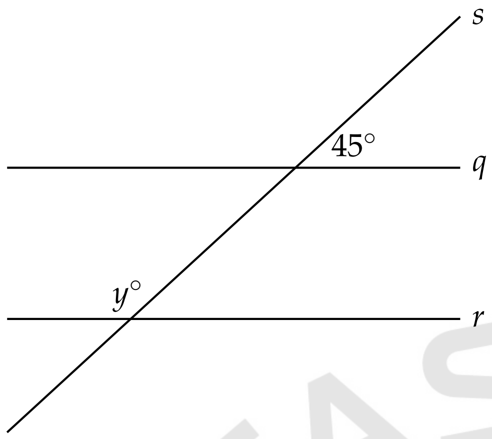
- Clockwise from top left, the 3 lines are labeled t, r, and s.
- Line t intersects both line r and line s.
- At the intersection of line t and line r, 1 angle is labeled clockwise from top left as follows:
  - Top right: a°
- At the intersection of line t and line s, 2 angles are labeled clockwise from top left as follows:
  - Top left: b°
  - Bottom left: c°
- Note: Figure not drawn to scale.

In the figure, lines r and s are parallel and line t intersects both lines. If $a = 4k + 57$ and $b = \frac{k}{2} + 55$, what is the value of c?

14. [October 2025]

In a right triangle, the measure of one of the acute angles is 14°. What is the measure of the other acute angle, in degrees? (Disregard the degree when entering your answer.)

15. [October 2025]



Note: Figure not drawn to scale.;

In the figure, line q is parallel to line r, and both lines are intersected by line s. If $y = 2x + 9$, what is the value of x?

A) 27

B) 73

C) 63

D) 18

16. [October 2025]

The measure of angle Z is 45°. What is the measure, in radians, of angle Z?

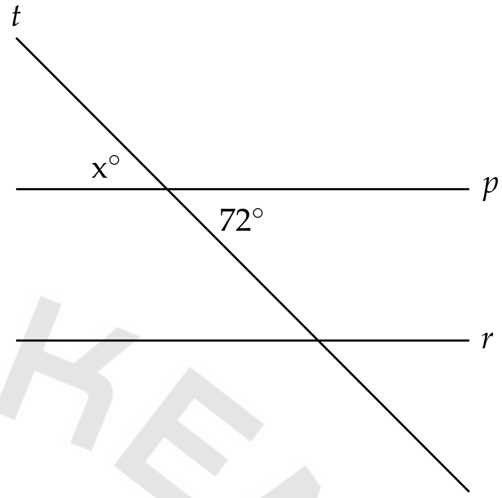
A) $\frac{1}{2}\pi$

B) $\frac{3}{4}\pi$

C) $\frac{1}{8}\pi$

D) $\frac{1}{4}\pi$

17. [October 2025]



Note: Figure not drawn to scale

In the figure, line p is parallel to line r,

and line t intersects both lines. What is the value of x?

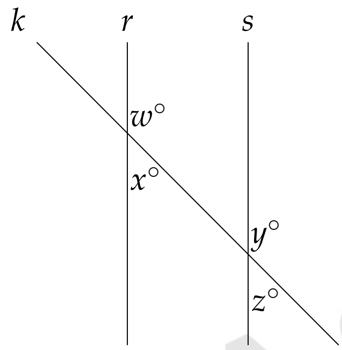
A) 36

B) 72

C) 180

D) 252

18. [October 2025]



Note: Figure not drawn to scale

In the figure shown, line k intersects lines r and s. If $w = 147$, which additional piece of information is sufficient to prove that lines r and s are parallel?

A) $x = 33$

B) $y = 147$

C) $w + y = 180$

D) $y + z = 180$

Answers: Lines and Angles

Number	Answer	Number	Answer
1	62	2	736
3	C	4	A
5	1360	6	B
7	A	8	C
9	B	10	D
11	C	12	A
13	117.4	14	76
15	C	16	D
17	B	18	B

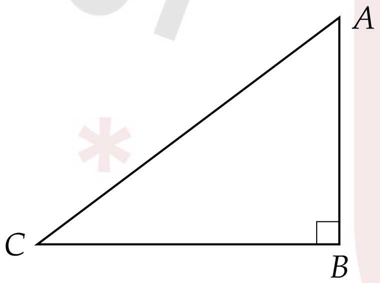
Triangles

1. [August 2025]

The length of each side of triangle A is 270 millimeters. The length of each side of triangle B is double the length of each side of triangle A. What is the length, in millimeters, of each side of triangle B?

A) 270
B) 272
C) 540
D) 1,080

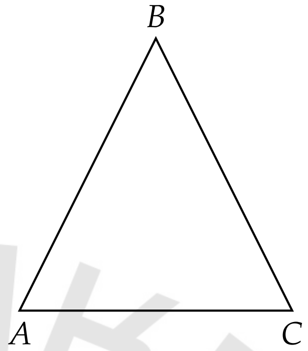
2. [August 2025]



In the right triangle shown, what is the value of $\cos A$?

A) $\frac{1}{47}$
B) $\frac{1}{29}$
C) $\frac{29}{47}$
D) $\frac{47}{29}$

3. [August 2025]



In the figure shown, the length of AB is 22. Which statement provides sufficient information to prove that triangle ABC is isosceles?

A) The length of AC is 11.
B) The length of BC is 22.
C) The measure of angle A is 70°.
D) The sum of the measures of angles A, B, and C is 180°.

4. [September 2025]

What is the y-intercept of the graph of $y = 5^x + 20$ in the xy-plane?

A) (0, 5)
B) (0, 21)
C) (0, 20)
D) (0, 25)

5. [September 2025]

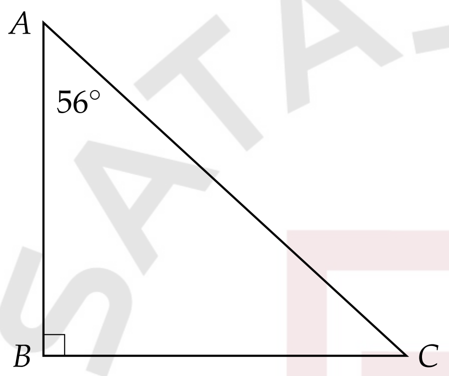
In triangle ABC, the measure of angle A is 56° and $AC = 35$. In triangle PQR, the measure of angle P is 56° and $PR = 105$. Which additional piece of information is sufficient to prove that triangle ABC is similar to triangle PQR?

A) $AB = 50$ and $PQ = 50$
B) The measures of angle B and angle R are 32° and 92°, respectively.
C) The measures of angle B and angle Q are 56° and 32°, respectively.
D) $AB = 50$ and $QR = 50$

6. [September 2025]

In triangle RST, angle T is a right angle, point L lies on RS, point K lies on ST, and LK is parallel to RT. If the length of RT is 72 units, the length of LK is 24 units, and the area of RST triangle is 396 square units, what is the length of KT, in units?

7. [October 2025]



Note: Figure not drawn to scale

- Angle B is a right angle.
- The measure of angle A is $56°$.
- Note: Figure not drawn to scale.

In right triangle ABC shown, what is the measure of $\angle C$?

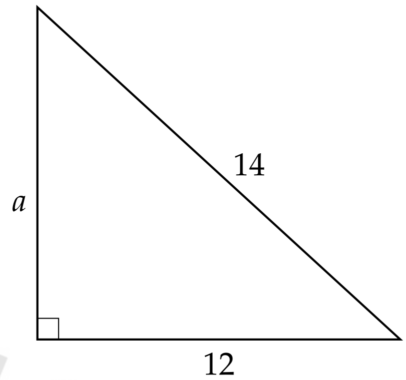
A) $124°$
B) $146°$
C) $56°$
D) $34°$

8. [October 2025]

A triangle has a base length of 48 centimeters and a height of 96 centimeters. What is the area, in square centimeters, of the triangle?

A) 2304
B) 1152
C) 4608
D) 144

9. [October 2025]

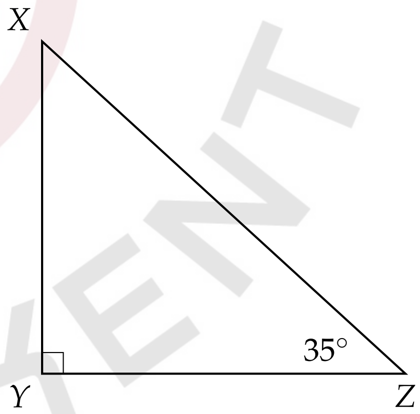


Note: Figure not drawn to scale

Which equation gives the relationship between the side length of the triangle shown?

A) $12^2 + 14^2 = a^2$
B) $a^2 + 12^2 = 14^2$
C) $a + 12 = 14$
D) $12 + 14 = a$

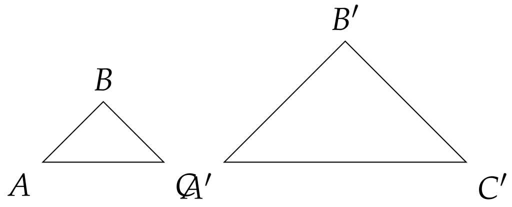
10. [October 2025]



Note: Figure not drawn to scale

In triangle XYZ, the measure of angle Y is 90 degrees. Triangle RST (not shown) is similar to triangle XYZ such that R, S, and T correspond to X, Y, and Z, respectively. Each side of triangle RST is 19 times the length of the corresponding side of triangle XYZ. What is the measure, in degrees, of one of the angles in triangle RST?

11. [October 2025]



Note: Figure not drawn to scale

Triangles ABC and $A'B'C'$ are shown. Triangle ABC is dilated by a scale factor of 6 to form triangle $A'B'C'$. If the length of $\overline{AB}$ is 18, what is the length of $\overline{A'B'}$?

A) 3
B) 6
C) 24
D) 108

12. [October 2025]

The table gives the areas and perimeters of two similar rectangles, where n is a constant

	Area (square inches)	Perimeter (inches)
Rectangle A	630	210
Rectangle B	2,520	n

What is the value of n?

A) 2100
B) 1680
C) 840
D) 420

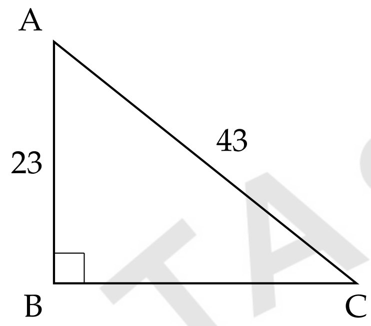
Answers: Triangles

Number	Answer	Number	Answer
1	C	2	C
3	B	4	B
5	B	6	22/3; 7.33
7	D	8	A
9	B	10	90
11	D	12	D

Trigonometry

1. [August 2025]

In the right triangle shown, what is the value o f $cosA$?



A) $\frac{1}{43}$

B) $\frac{1}{23}$

C) $\frac{23}{43}$

D) $\frac{43}{23}$

2. [August 2025]

$$RS = 504$$
$$ST = 128$$
$$TR = 520$$

The side lengths of triangle RST are given. Triangle RST is similar to triangle UVW, where S corresponds to V and T corresponds to W. What is the value of tan W?

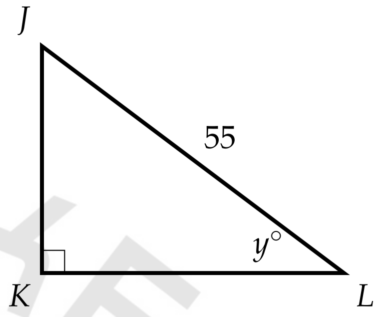
A) $\frac{16}{65}$

B) $\frac{16}{63}$

C) $\frac{63}{65}$

D) $\frac{63}{16}$

3. [September 2025]



Note: Figure not drawn to scale.

In triangle JKL, $\cos y^\circ = \dfrac{54}{55}$. What is the length of KL?

4. [September 2025]

Quadrilateral $MNOP$ is similar to quadrilateral $QRST$ such that $M, N, O,$ and P correspond to $Q, R, S,$ and T, respectively. The length of each side of quadrilateral $MNOP$ is 2 times the length of its corresponding side in quadrilateral $QRST$. The measure of angle P is 50° and $ST = 8$. Which of the following must be true?

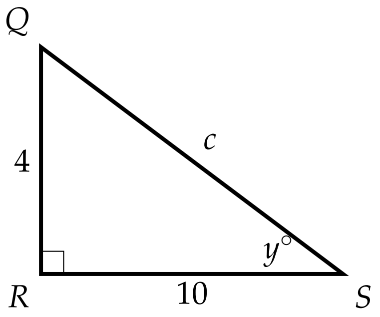
A) $OP = 8$

B) The measure of angle T is 50°.

C) The measure of angle R is 100°.

D) $OP = 32$

5. [September 2025]



Note: Figure not drawn to scale.

In triangle QRS, what is the value of $\tan y°$?

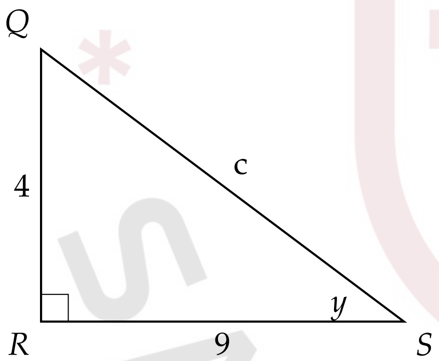
A) $\frac{c}{4}$

B) $\frac{c}{10}$

C) $\frac{4}{10}$

D) $\frac{10}{4}$

6. [September 2025]



In triangle QRS, what is the value of tan(y)?

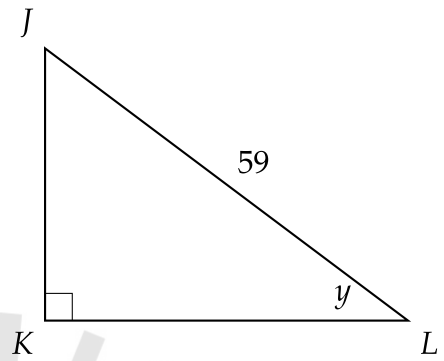
A) $\frac{c}{9}$

B) $\frac{4}{9}$

C) $\frac{c}{4}$

D) $\frac{9}{4}$

7. [September 2025]



In triangle JKL, $cos(y) = \frac{58}{59}$. What is the length of KL?

8. [October 2025]

Triangle ABC is similar to triangle DEF, where angle A corresponds to angle D and angle C corresponds to angle F. Angles C and F are right angles. If $\tan A = \dfrac{612}{35}$, what is the value of $\sin D$?

A) $\frac{612}{613}$

B) $\frac{35}{613}$

C) $\frac{613}{612}$

D) $\frac{613}{35}$

9. [October 2025]

Triangle ABC is similar to triangle DEF, where angle A corresponds to angle D and angle C corresponds to angle F. Angles B and E are right angles. If $\cos A = \dfrac{14}{\sqrt{421}}$, what is the value of $\dfrac{1}{sinF}$

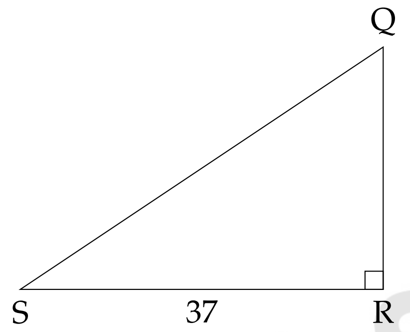
A) $\frac{15}{\sqrt{421}}$

B) $\frac{14}{\sqrt{421}}$

C) $\frac{\sqrt{421}}{14}$

D) $\frac{\sqrt{421}}{15}$

10. [October 2025]



Note: Figure not drawn to scale

In triangle QRS shown, $QR < RS$. Which expression represents the length of QS?

A) $37 \cos Q$

B) $37 \sin Q$

C) $\frac{37}{\cos Q}$

D) $\frac{37}{\sin Q}$

Answers: Trigonometry

Number	Answer	Number	Answer
1	C	2	D
3	54	4	B
5	C	6	B
7	58	8	A
9	C	10	C

Circle

1. [September2025]

A circle has circumference of 89π centimeters. What is the diameter, in centimeter, of the circle?

2. [September2025]

Circles A and B are graphed in the xy-plane. Circle A is represented by the equation $(x+8)^2+(y-8)^2=64$ and intersects the x-axis at the point (r,s). Circle B has its center at (r,s) and has a radius of the same length as circle A. Which equation represents circle B?

A) $x^2+(y+8)^2=64$
B) $x^2+(y-8)^2=64$
C) $(x+8)^2+y^2=64$
D) $(x-8)^2+y^2=64$

3. [August 2025]

Circle N has a radius of 8 millimeters (mm). Circle M has an area of 144π mm². What is the total area, in mm², of circles N and M?

A) 20π
B) 160π
C) 176π
D) 208π

4. [August 2025]

Points F and G lie on a circle with center H. Segment FG is a diameter of the circle. If the length of segment FG is 82 centimeters, what is the length, in centimeters, of segment FH?

5. [August 2025]

A circle in the xy-plane has its center at (-4, -7). Line k is tangent to this circle at the point (-7, -8). What is the slope of line k?

A) -3
B) $-\frac{1}{3}$
C) $\frac{1}{3}$
D) 3

6. [September 2025]

A circle in the xy-plane has its center at (2, 9). Line t is tangent to this circle at the point $(a,-4)$, where a is a constant. The slope of line t is $\frac{6}{5}$. What is the value of a?

A) $-\frac{68}{5}$
B) $-\frac{53}{6}$
C) $\frac{77}{6}$
D) $\frac{88}{5}$

7. [August 2025]

Points F and G lie on a circle with center H. Segment FG is a diameter of the circle. If the length of segment FG is 98 centimeters, what is the length, in centimeters, of segment FH?

8. [September 2025]

In the xy-plane, the graph of the equation $(x-3)^2+(y-8)^2 = 9$ is a circle. The point $(6,c)$, where c is a constant, lies on this circle. What is the value of c?

9. [September 2025]

In the xy-plane, an equation of circle A is $(x-2)^2+(y-3)^2=25$. Circle B has the same center as circle A but has a radius that is twice the radius of circle A. Which equation represents circle B?

A) $(x-2)^2+(y-3)^2=625$
B) $(x-2)^2+(y-3)^2=50$
C) $(x-2)^2+(y-3)^2=250(x-2)^2+(y-3)^2=100$
D) $(x-2)^2+(y-3)^2=100$

10. [September 2025]

In the xy-plane, the graph of the equation $(x-3)^2+(y-8)^2=9$ is a circle. The point $(6,c)$, where c is a constant, lies on this circle. What is the value of c?

11. [September 2025]

In the xy-plane, an equation of circle A is $(x-2)^2+(y-4)^2=9$. Circle B has the same center as circle A but has a radius that is twice the radius of circle A. Which equation represents circle B?

A) $(x-2)^2+(y-4)^2=81$
B) $(x-2)^2+(y-4)^2=36$
C) $(x-2)^2+(y-4)^2=18$
D) $(x-2)^2+(y-4)^2=54$

12. [October 2025]

Circle K has a radius of 8 millimeters (mm). Circle L has an area of 144 mm^2. What is the total area, in mm^2, of circles K and L?

A) 40π
B) 20π
C) 80π
D) 208π

13. [October 2025]

In the xy-plane, the graph of the equation $(x-3)^2+(y-8)^2=9$ is a circle. The point $(6,c)$, where c is a constant, lies on this circle. What is the value of c?

14. [October 2025]

In the xy-plane, an equation of circle A is $(x-2)^2+(y-4)^2=9$. Circle B has the same center as circle A but has a radius that is twice the radius of circle A. Which equation represents circle B?

A) $(x-2)^2+(y-4)^2=81$
B) $(x-2)^2+(y-4)^2=36$
C) $(x-2)^2+(y-4)^2=18$
D) $(x-2)^2+(y-4)^2=54$

15. [October 2025]

A circle in the xy-plane has its center at $(-4,1)$ and the point $(-8,4)$ lies on the circle. Which equation represents this circle?

A) $(x-4)^2+(y+1)^2=5$
B) $(x+4)^2+(y-1)^2=25$
C) $(x-4)^2+(y+1)^2=25$
D) $(x+4)^2+(y-1)^2=5$

16. [October 2025]

A circle has center P, and points A and B lie on the circle. The measure of arc AB is $45°$ and the length of arc AB is 4π units. What is the length, in units, of the radius of the circle?

Answers: Circle

Number	Answer	Number	Answer
1	89	2	C
3	D	4	41
5	A	6	D
7	49	8	8
9	D	10	8
11	B	12	D
13	8	14	B
15	B	16	16

Area and Volume

1. [September 2025]

A right circular cone has a base diameter of 28 inches and a height of 6 inches. The volume of this cone is $k\pi$ cubic inches. What is the value of k?

2. [September 2025]

A rectangular poster has an area of 330 square inches. A copy of poster is made in which the length and width of the original poster are each increased by 50%. What is the area of the copy, in square inches?

3. [September 2025]

A piece of string with a length of 108 inches is cut into two parts. One part has a length of x inches, and the other part has a length of y inches. The value of x is 8 more than 4 times the value of y. What is the value of x?

A) 25

B) 29

C) 58

D) 88

4. [September 2025]

A rectangle has a length that is 73 times its width. The function $y = (73w)(w)$ represents this situation, where y is the area, in square feet, of the rectangle and $y > 0$. Which of the following is the best interpretation of $73w$ in this context?

A) The length of the rectangle, in feet.

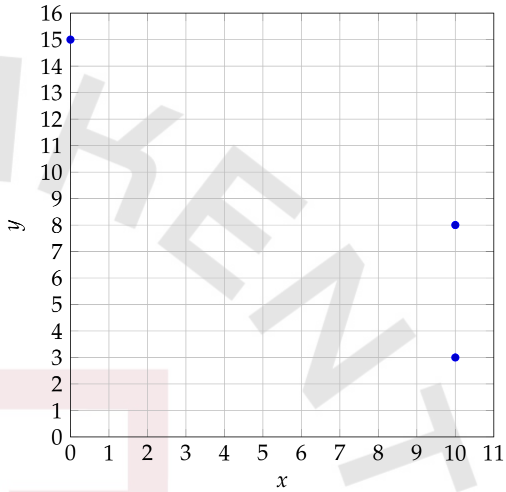
B) The area of the rectangle, in square feet.

C) The difference between the length and the width of the rectangle, in feet.

D) The width of the rectangle, in feet.

5. [August 2025]

The points (0, 15), (10, 8), and (10, 3) are shown in the xy-plane, where the x-axis and y-axis are measured in units.



These points define one of the bases of a triangular prism. The distance between the two bases of the prism is 20 units. What is the volume, in cubic units, of the prism?

6. [September 2025]

Rectangles $ABCD$ and $EFGH$ are similar. The length of each side of $EFGH$ is 6 times the length of the corresponding side of $ABCD$. The area of ABCD is 90 square units. What is the area, in square units, of $EFGH$?

A) 15

B) 36

C) 540

D) 3,240

7. [October 2025]

The width of a rectangle is 3 centimeters. The length of the rectangle is 20 centimeters longer than the width. What is the area, in square centimeters, of this rectangle?

A) 6

B) 26

C) 69

D) 3

8. [October 2025]

$$h(x) = \frac{1}{2}(x)(77)$$

The given function h represents the area, in square inches, of a triangle with a base of x inches and a fixed height. What is the height, in inches, of the triangle?

A) $\frac{1}{77}$

B) $\frac{77}{2}$

C) 77

D) 154

9. [October 2025]

Cylinder	Volume (cubic units)
Right circular cylinder A	32π
Right circular cylinder B	4000π

The table shows the volume of two similar solids, right circular cylinder A and right circular cylinder B. The radius of the right circular cylinder A is 4 units. The surface area of right circular cylinder A is $k\pi$ square units, and the surface area of right circular cylinder B is $n\pi$ square units, where k and n are constants. What is the value of $n - k$? (The surface area of a right circular cylinder with radius r and height h is $2\pi r^2 + 2\pi rh$.)

10. [October 2025]

Cube B has a volume of 216 cubic inches. The length of each edge of cube A is k times the length of each edge of cube B. If cube A has a volume of 5,832 cubic inches, what is the value of k?

A) 27

B) 3

C) 6

D) 36

11. [September 2025]

A rectangle has an area of 48 square meters and a length of 12 meters. What is the width, in meters, of the rectangle?

A) 576

B) 36

C) 4

D) 144

12. [September 2025]

The length of the diagonal of a square is $\dfrac{196\sqrt{2}}{2}$ units. What is the area, in square units, of the square?

A) $19,208$

B) 98

C) 392

D) $9,604$

13. [September 2025]

In triangle RST, angle T is a right angle, point L lies on RS, point K lies on ST, and LK is parallel to RT. If the length RT is 63 units, the length of LK is 21 units, and the area of triangle RST is 630 square units, what is the length of KT, in units?

14. [September 2025]

A right square pyramid has a height of 6 units and a volume of 128 cubic units. What is the length, in units, of one side of the base of the pyramid?

A) 256

B) 8

C) $\frac{64}{3}$

D) $\frac{8\sqrt{3}}{3}$

15. [September 2025]

A rectangle has an area of 66 square meters and a length of 11 meters. What is the width, in meters, of the rectangle?

A) 6

B) 55

C) 726

D) 121

16. [September 2025]

The length of the diagonal of a square is $\frac{186\sqrt{2}}{2}$ units. What is the area, in square units, of the square?

A) $8,649$

B) $17,298$

C) 93

D) 372

17. [September 2025]

Cube B has a volume of 125 cubic inches. The length of each edge of cube A is k times the length of each edge of cube B. If cube A has a volume of 3,375 cubic inches, what is the value of k?

A) 27

B) 3

C) 25

D) 5

18. [September 2025]

A right square pyramid has a height of 6 units and a volume of 128 cubic units. What is the length, in units, of one side of the base of the pyramid?

A) 8

B) 256

C) $\frac{64}{3}$

D) $\frac{8\sqrt{3}}{3}$

19. [September 2025]

Circle K has a radius of 8 millimeters (mm). Circle L has an area of 144 mm^2. What is the total area, in mm^2, of circles K and L?

A) 40π

B) 20π

C) 80π

D) 208π

20. [September 2025]

A triangle has a base length of 48 centimeters and a height of 96 centimeters. What is the area, in square centimeters, of the triangle?

A) 2304

B) 1152

C) 4608

D) 144

21. [October 2025]

A right circular cylinder has a height of 16 inches and a volume of 3136π cubic inches. What is the radius, in inches, of this cylinder?

22. [October 2025]

The length of a side of square X is 9 centimeters. The area of rectangle Y is 32 square centimeters. What is the total area, in square centimeters, of square X and rectangle Y?

A) 145

B) 113

C) 82

D) 81

23. [September 2025]

The density of a certain type of marble stone is 2.6000 grams per cubic centimeter. If a sample of this type of stone is in the shape of a sphere with a diameter of 31.000 centimeters, what is the mass of this sample, in grams, to the nearest whole number? (Use 3.14159 for π.)

Answers: Area and Volume

Number	Answer	Number	Answer
1	392	2	742.5
3	D	4	A
5	500	6	D
7	C	8	C
9	1152	10	B
11	C	12	D
13	40/3; 13.33	14	B
15	A	16	A
17	B	18	A
19	D	20	A
21	14	22	B
23	40556		

Acknowledgements

We've reached the end of "MathBook 3.0 by @satashkent's R&D Team". Before we close, let's acknowledge the dedicated team that brought this project to life.

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Every question in this book has been carefully selected, entered, and reviewed to ensure accuracy and exam-level quality.

Powered by **@satashkent**'s Research and Development Team.

SATASHKENT
College Prep Community

Many real exam questions are officially and unofficially shared across the Internet, providing valuable practice materials for SAT students. However, these resources are often scattered, unstructured, and not categorized by topic, making it difficult for students to focus on specific areas of improvement.

Recognizing this challenge, our team created *MathBook*—a structured collection of *real exam questions from the past years, categorized by topic.* This book is designed for *all SAT learners* who want an efficient and organized approach to mastering Digital SAT Math. We have dedicated countless hours to compiling and structuring these questions to ensure you get the most relevant and targeted practice.

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